

LAPORAN PRAKTIKUM
POSTTEST 6
ALGORITMA PEMROGRAMAN LANJUT

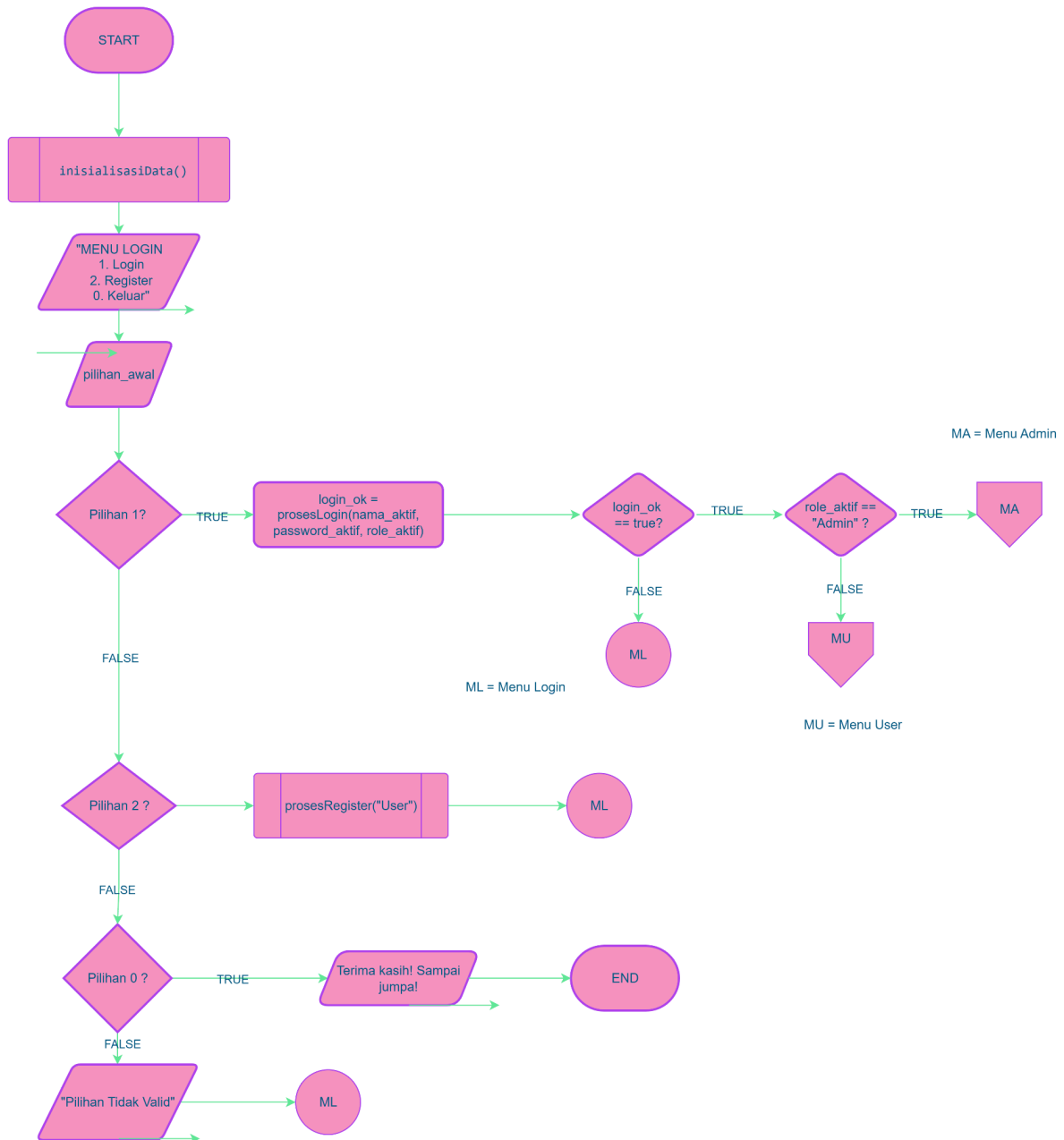


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A2 '25

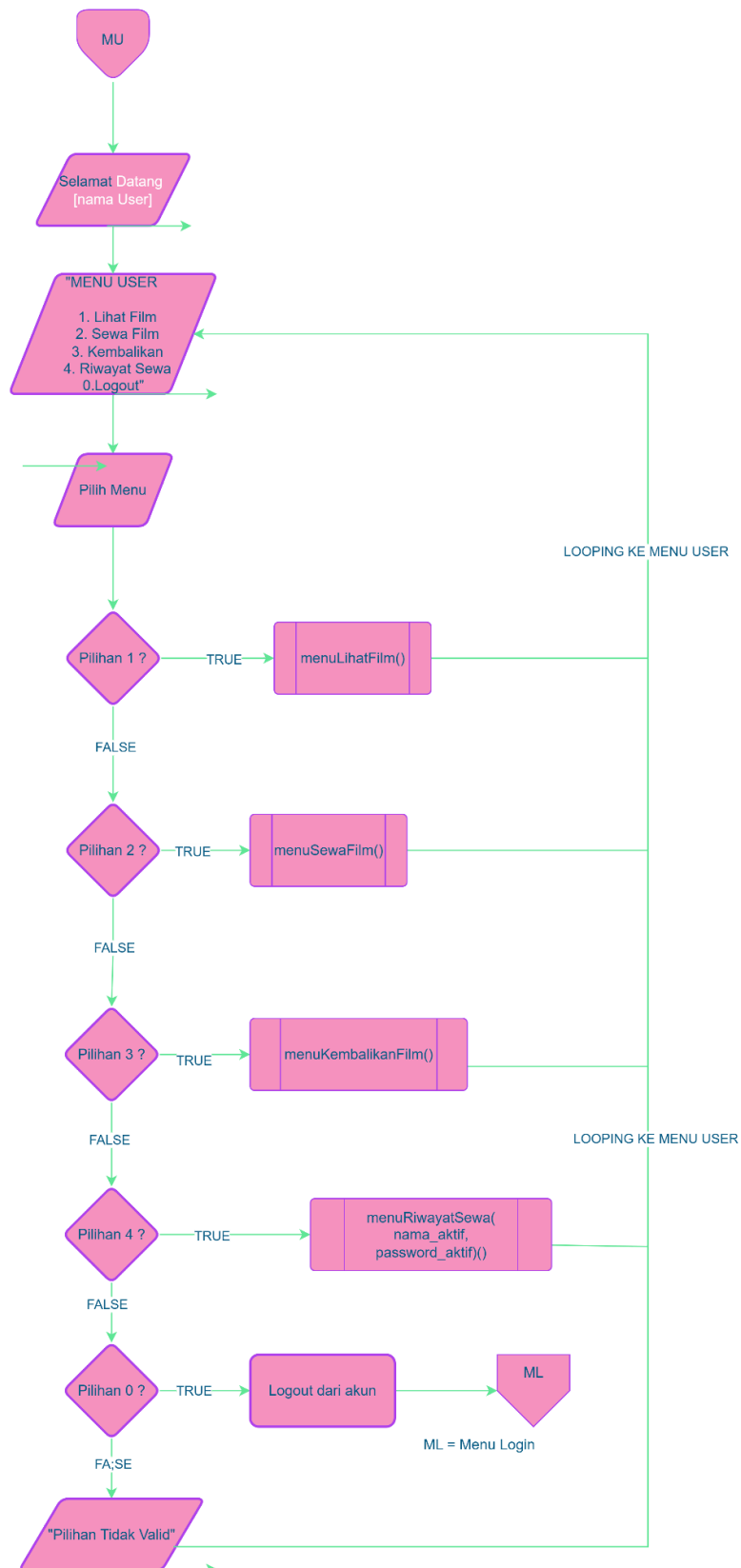
PROGRAM STUDI INFORMATIKA
UNIVERSITAS MULAWARMAN
SAMARINDA
2026

1. Flowchart

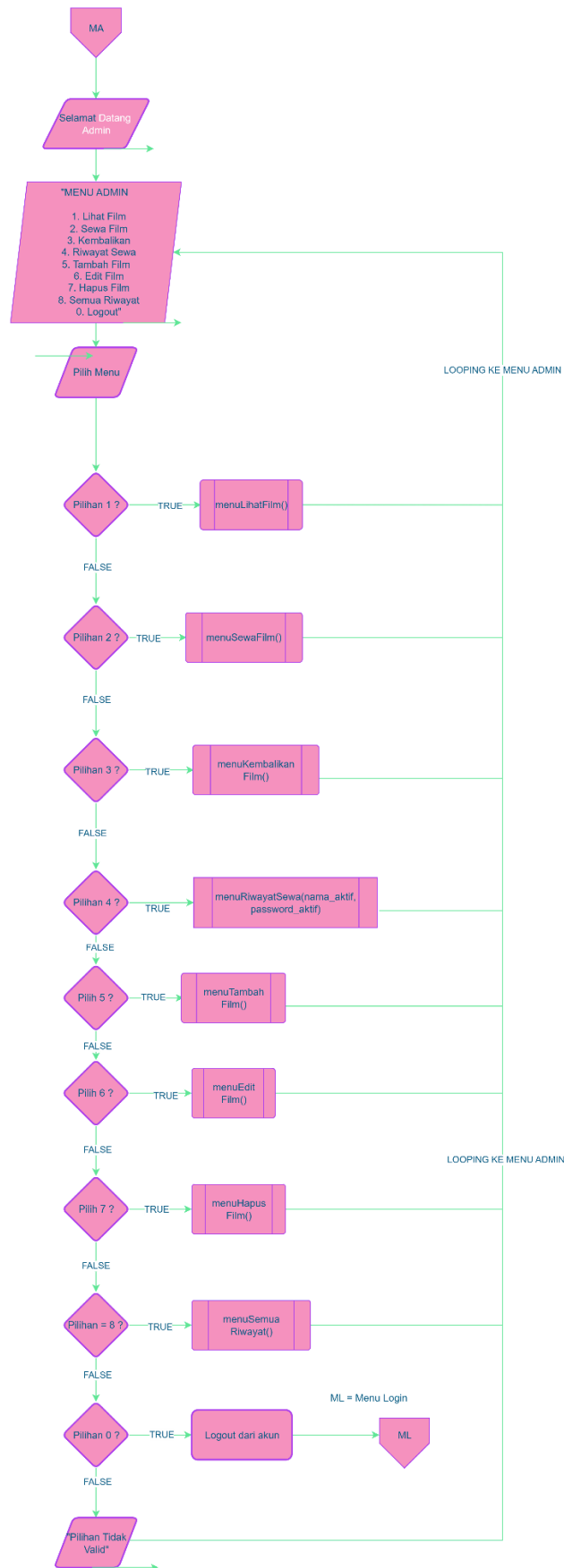
1.1 Gambar Bagian-Bagian Flowchart



Gambar 1.1 Flowchart Bagian Menu Login

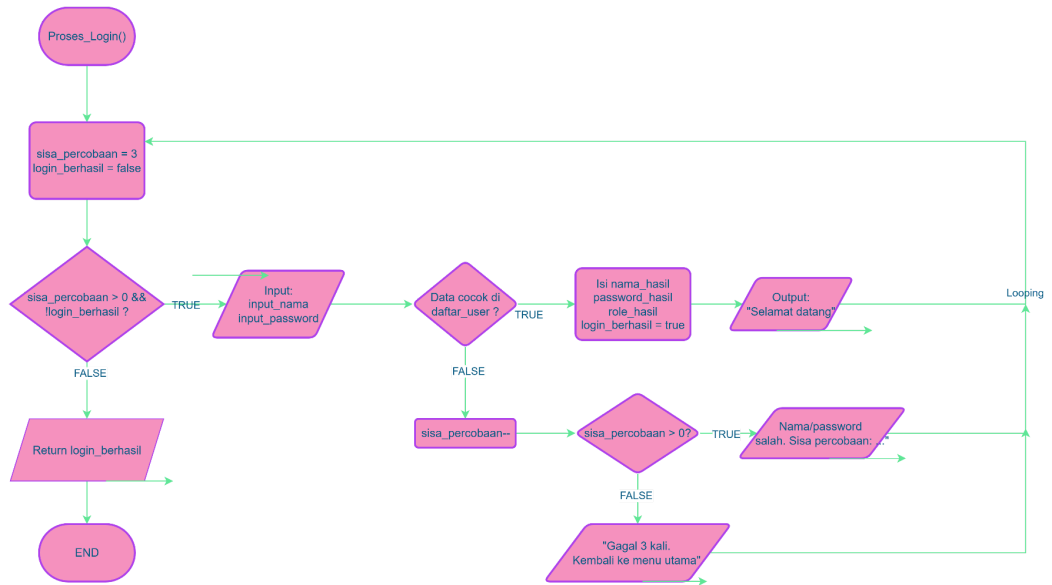


Gambar 1.2 Flowchart Bagian Menu User



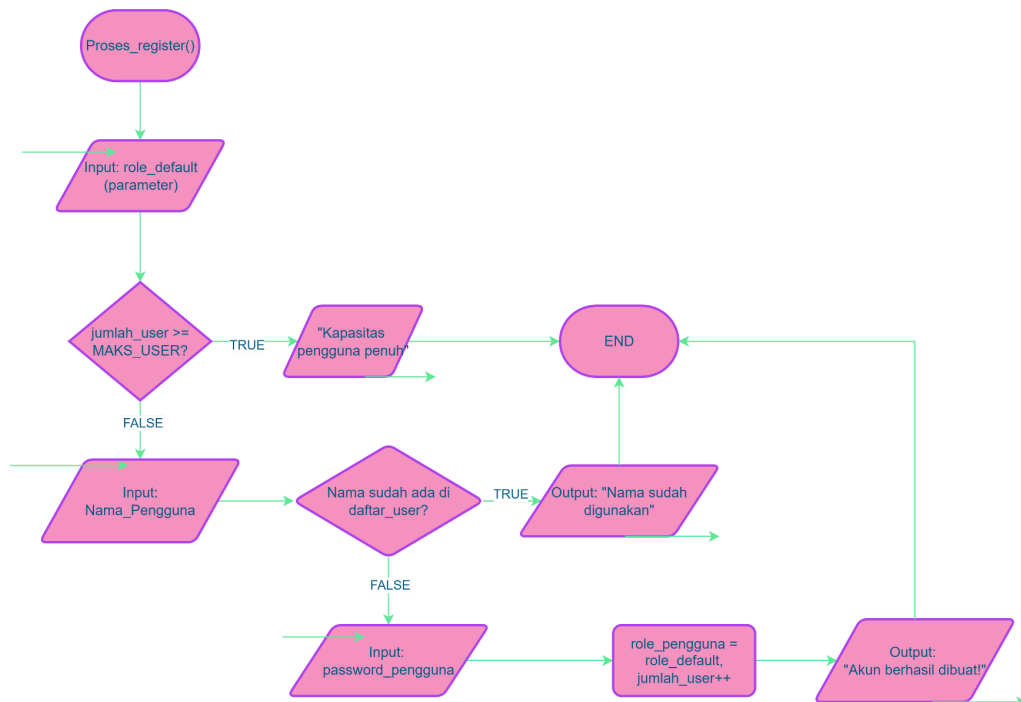
Gambar 1.3 Flowchart Bagian Menu Admin

PROSES
LOGIN

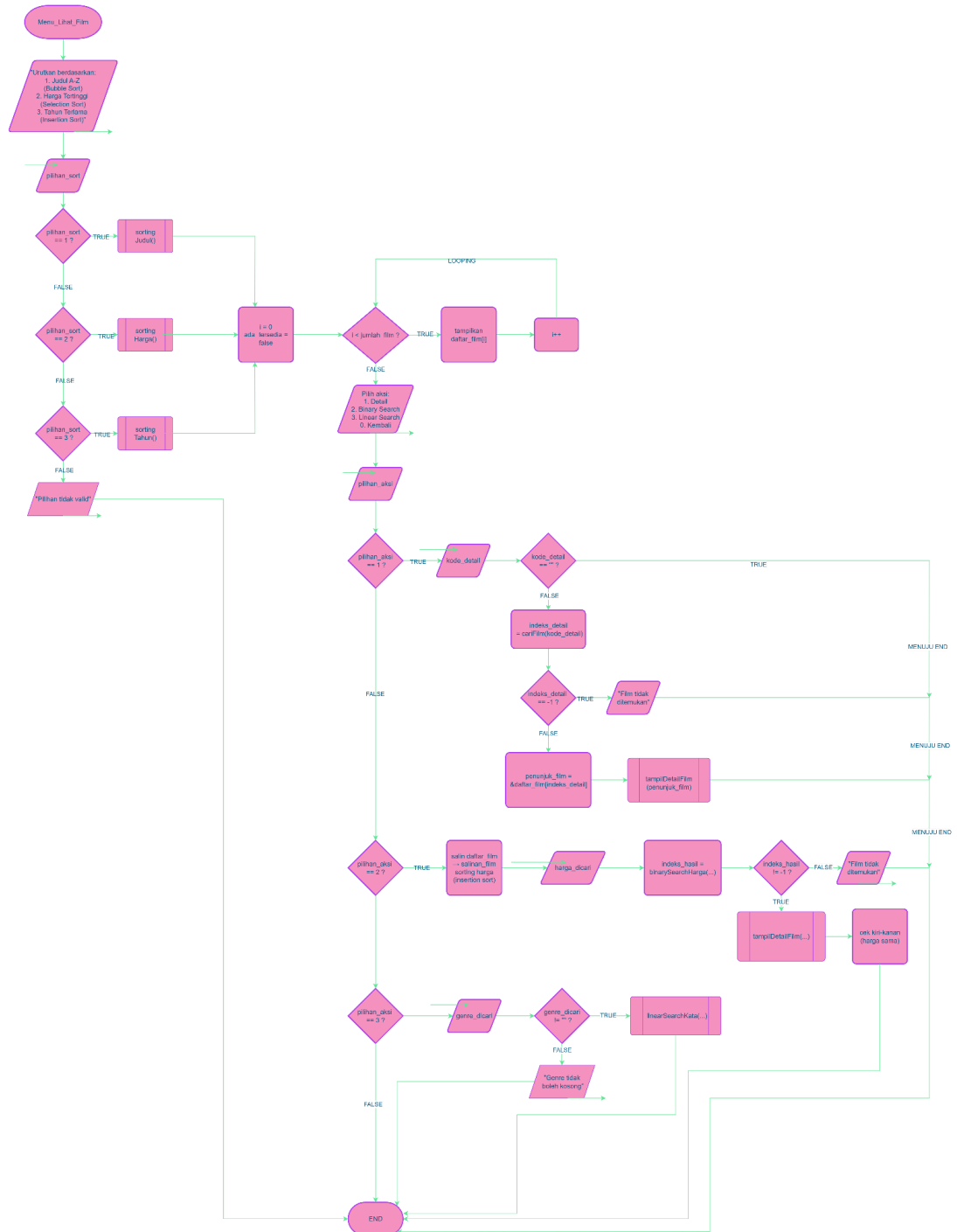


Gambar 1.4 Proses Login

PROSES
REGISTER

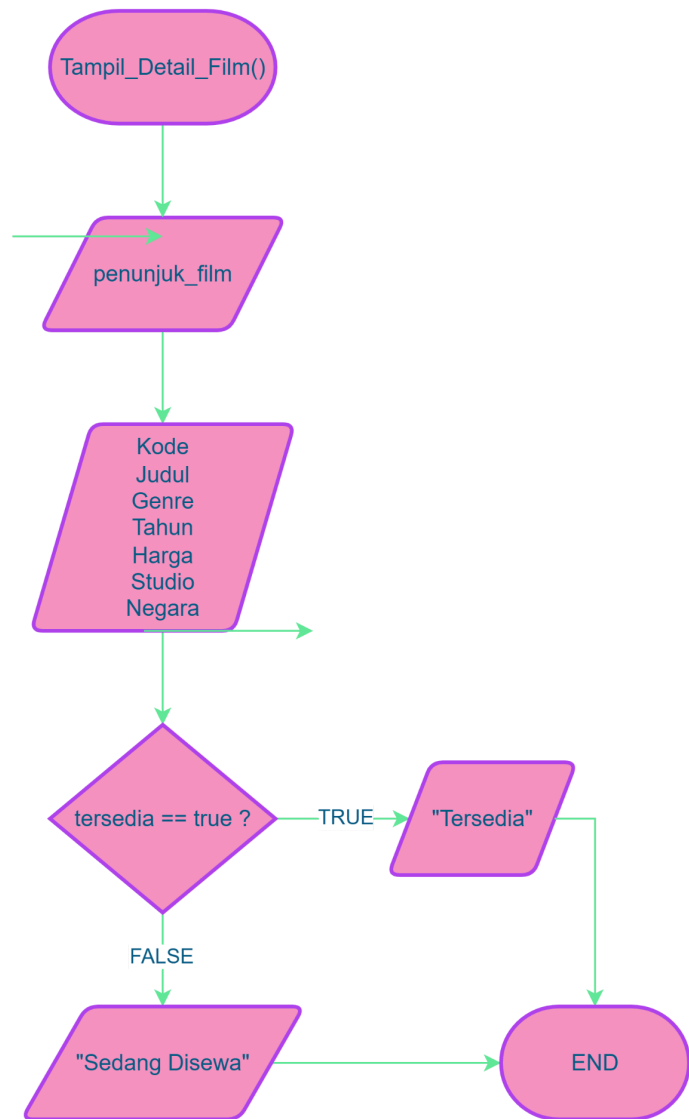


Gambar 1.5 Proses Register



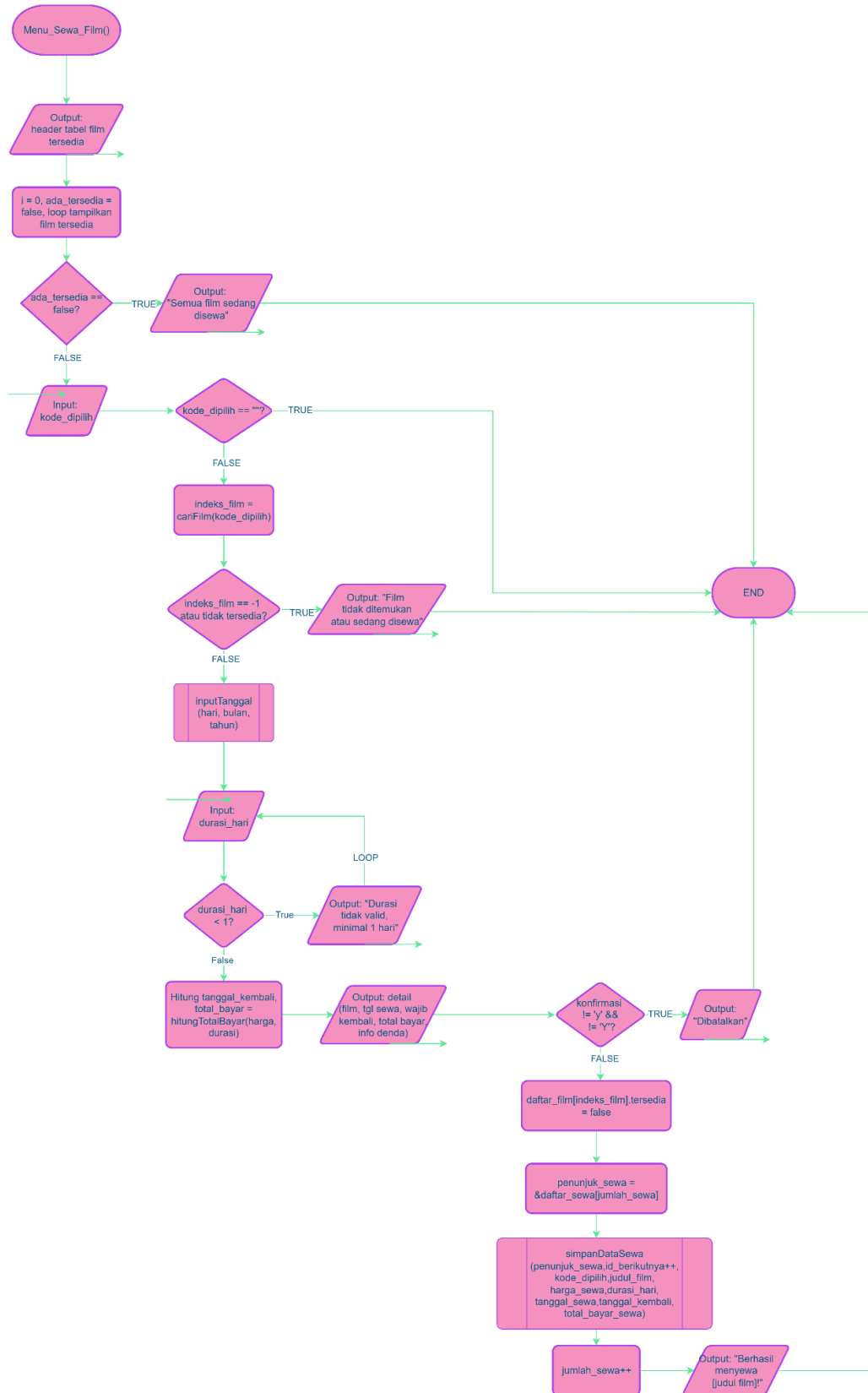
Gambar 1.6 Menu Lihat Film

TAMPIL DETAIL FILM

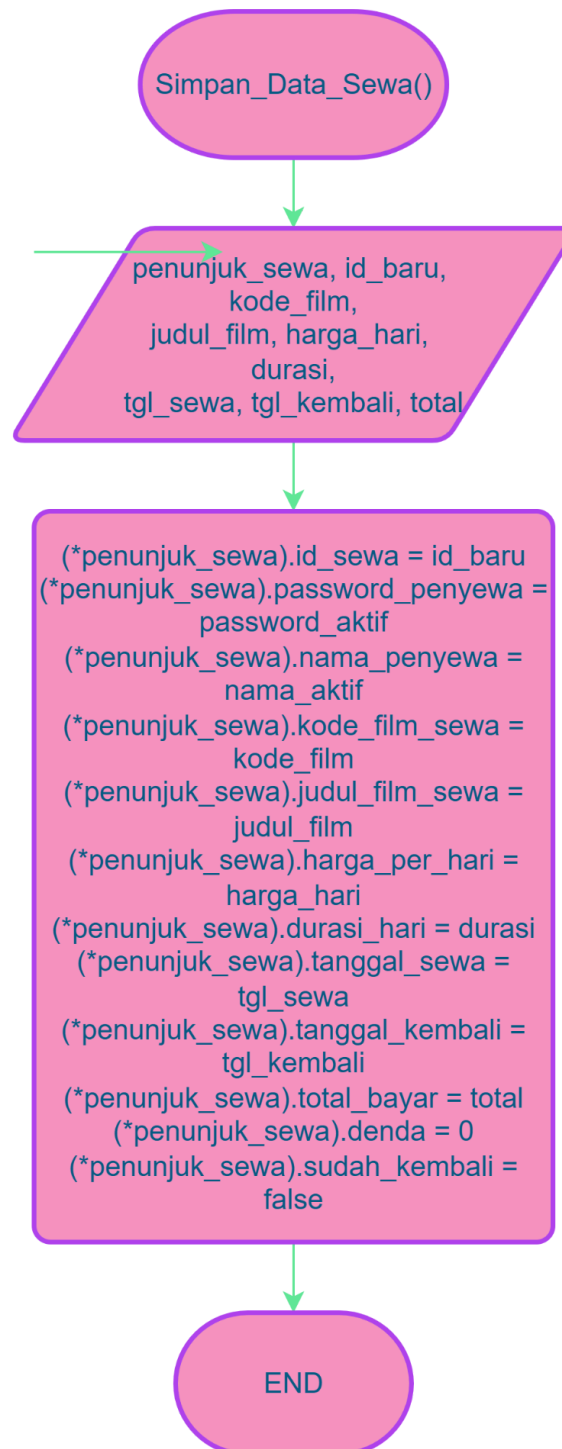


Gambar 1.7 Tampil Detail Film

menu sewa FILM

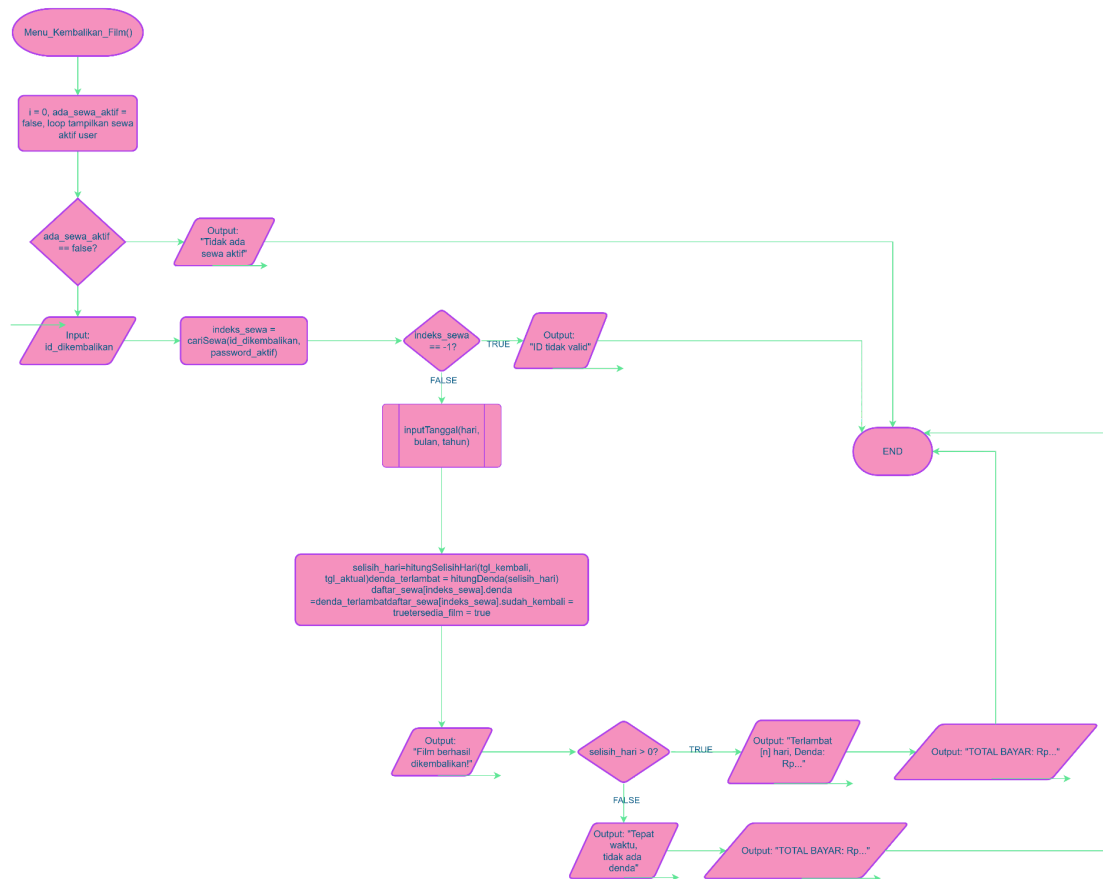


Gambar 1.8 Menu Sewa Film



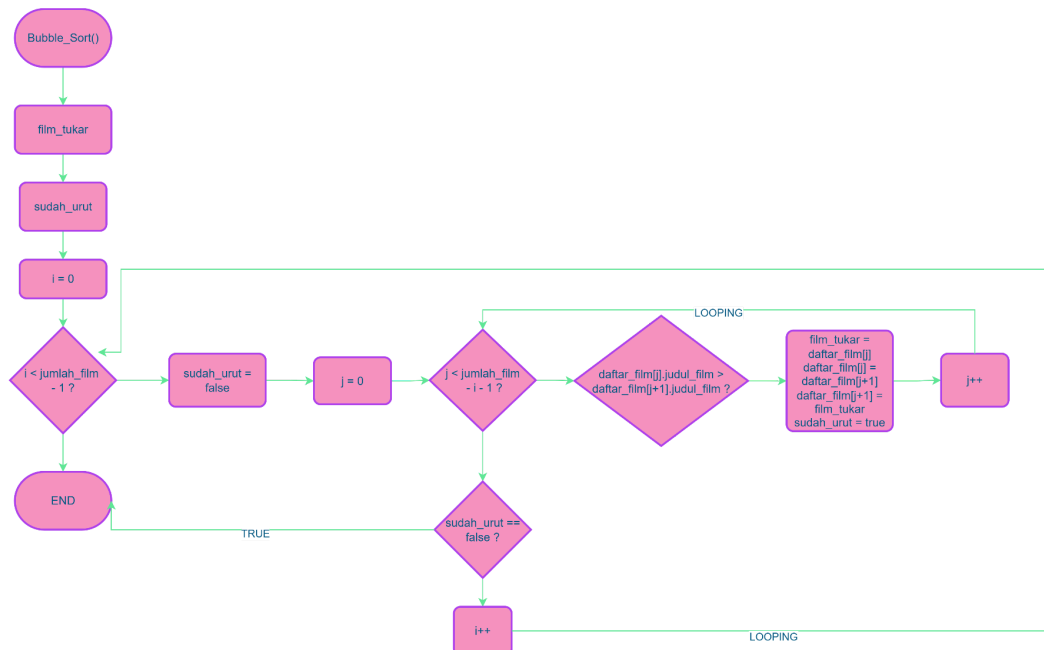
Gambar 1.9 Simpan Data Sewa

MENU
KEMBALIKAM
FILM

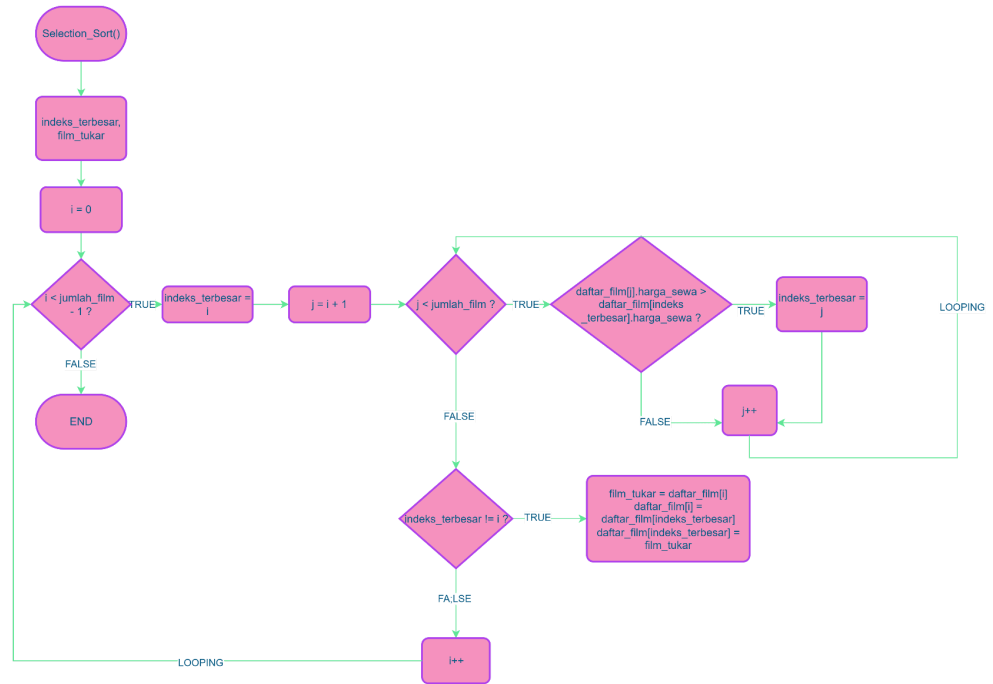


Gambar 1.10 Menu Kembalikan Film

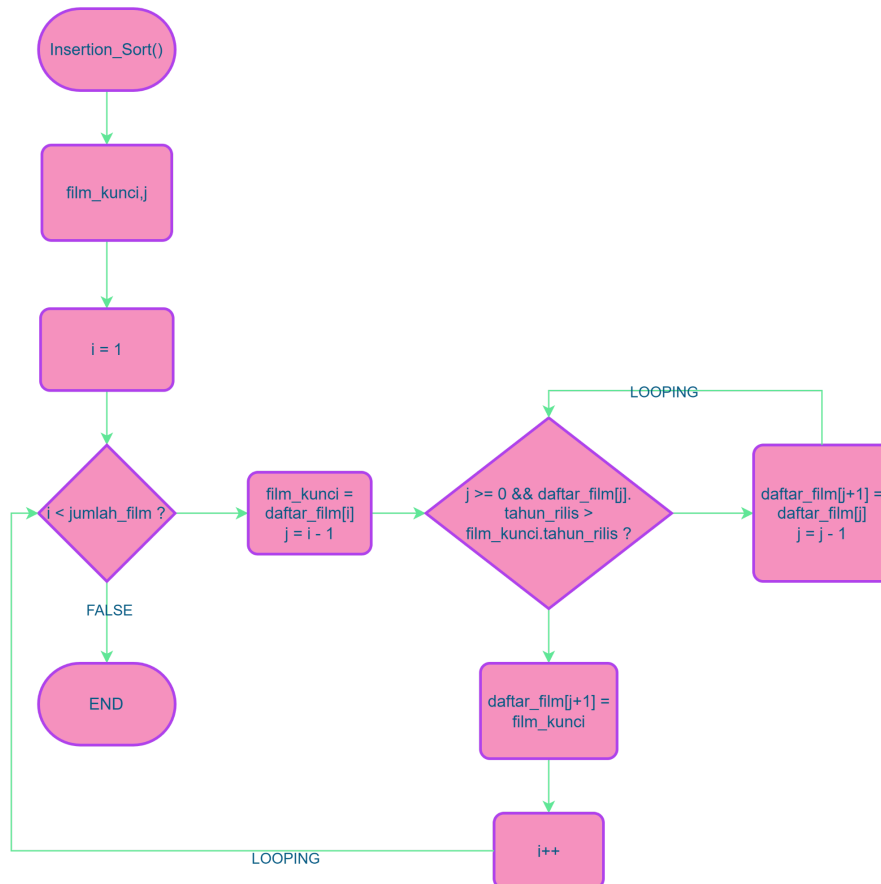
BUBBLE
SORT



Gambar 1.11 Sorting Judul

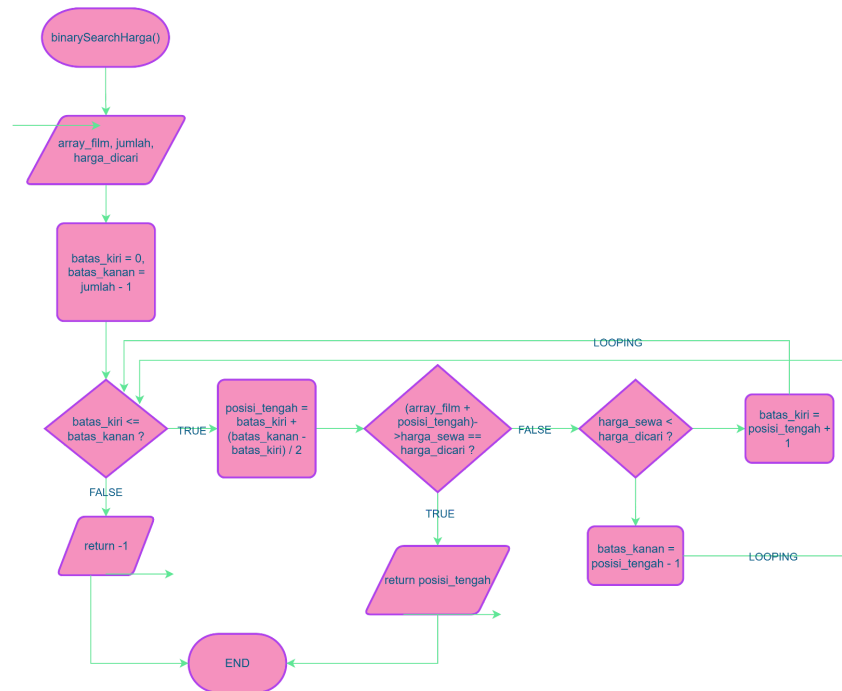


Gambar 1.12 Sorting Harga



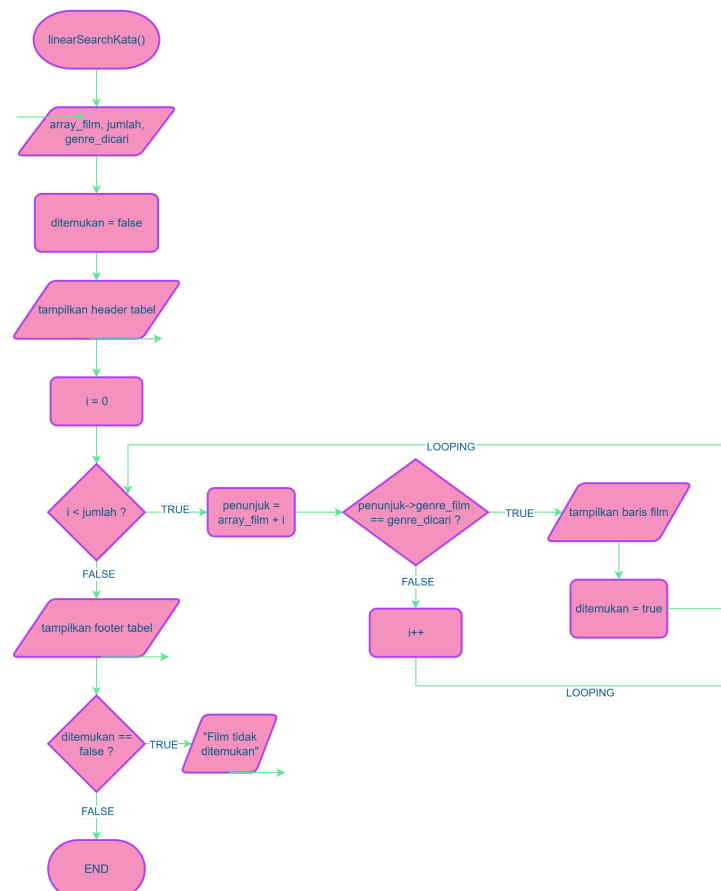
Gambar 1.13 Sorting Tahun

BINARY SEARCH



Gambar 1.14 Searching Harga

LINEAR SEARCH



Gambar 1.15 Searching Genre

1.2 Penjelasan Flowchart

Program CinemaRent diawali dengan inisialisasi data film dan pengguna, kemudian menampilkan menu utama dengan pilihan login, register, dan keluar. Proses login ditangani oleh fungsi `prosesLogin()` dengan batas tiga kali percobaan menggunakan pass by reference, dan jika berhasil pengguna diarahkan ke menu sesuai role yaitu Menu User atau Menu Admin. Menu User menyediakan empat fitur yaitu lihat film, sewa film, kembalikan film, dan riwayat sewa, sedangkan Menu Admin memiliki fitur tambahan berupa tambah, edit, hapus film, dan semua riwayat. Pada fitur Lihat Film, pengguna terlebih dahulu memilih metode sorting antara Bubble Sort untuk judul A-Z, Selection Sort untuk harga tertinggi, atau Insertion Sort untuk tahun terlama, setelah tabel ditampilkan pengguna dapat memilih aksi lanjutan yaitu melihat detail film by kode, mencari film by harga menggunakan Binary Search, atau mencari film by genre menggunakan Linear Search. Fungsi `binarySearchHarga()` menerima parameter pointer `Film*` dan bekerja dengan membagi array yang sudah terurut menjadi dua bagian di setiap iterasi untuk menemukan film berdasarkan harga, sedangkan fungsi `linearSearchKata()` menerima parameter pointer `Film*` dan menelusuri seluruh array satu per satu untuk menampilkan semua film yang genrenya sesuai dengan input pengguna. Fitur sewa menggunakan `inputTanggal()` dengan pass by reference untuk mencatat tanggal sewa dan menghitung tanggal wajib kembali, lalu data disimpan melalui prosedur `simpanDataSewa()` menggunakan pointer `DataSewa*`. Fitur kembalikan film menghitung selisih hari keterlambatan dan denda menggunakan fungsi rekursif `hitungDendaRekursif()`, serta menerapkan function overloading pada `hitungDenda()`. Dalam flowchart digunakan simbol terminator untuk START dan END, jajaran genjang untuk input dan output, belah ketupat untuk percabangan, kotak biasa untuk proses, dan predefined process untuk setiap pemanggilan fungsi atau prosedur.

2. Deskripsi Singkat Program

Program CinemaRent merupakan aplikasi berbasis console menggunakan bahasa C++ untuk mengelola penyewaan film dengan dua jenis pengguna yaitu User dan Admin. User dapat melihat daftar film, menyewa, mengembalikan, dan melihat riwayat sewa, sedangkan Admin memiliki fitur tambahan untuk mengelola data film dan melihat seluruh riwayat transaksi. Pada Posttest 6, program dikembangkan dengan menambahkan dua metode searching yaitu Binary Search pada fungsi `binarySearchHarga()` untuk mencari film berdasarkan harga (angka) dengan syarat data sudah terurut, dan Linear Search pada fungsi `linearSearchKata()` untuk mencari film berdasarkan genre (kata) dengan menelusuri seluruh data satu per satu, di mana kedua fungsi tersebut menggunakan pointer `Film*` sebagai parameter sehingga memenuhi poin plus yang disyaratkan.

3. Source Code

3.1 Deklarasi Struct & Variabel Global

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

struct Studio {
    string nama_studio;
    string negara_studio;
};

struct Film {
    string kode_film;
    string judul_film;
    string genre_film;
    int    tahun_rilis;
    int    harga_sewa;
    Studio studio;
    bool   tersedia;
};

struct Pengguna {
    string nama_pengguna;
    string password_pengguna;
    string role_pengguna;
};

struct Tanggal {
    int hari;
    int bulan;
    int tahun;
};

struct DataSewa {
    int    id_sewa;
    string password_penyewa;
    string nama_penyewa;
    string kode_film_sewa;
    string judul_film_sewa;
    int    harga_per_hari;
    int    durasi_hari;
    Tanggal tanggal_sewa;
    Tanggal tanggal_kembali;
    int    total_bayar;
    int    denda;
```

```

    bool    sudah_kembali;
};

#define MAKS_FILM      50
#define MAKS_USER      20
#define MAKS_SEWA      100
#define DENDA_PER_HARI 5000

Film    daftar_film[MAKS_FILM];
int     jumlah_film = 0;

Pengguna daftar_user[MAKS_USER];
int     jumlah_user = 0;

DataSewa daftar_sewa[MAKS_SEWA];
int     jumlah_sewa    = 0;
int     id_berikutnya  = 1;

string nama_aktif      = "";
string password_aktif  = "";
string role_aktif      = "";

```

3.2 Deklarasi Fungsi & Prosedur

```

void inisialisasiData();
void menuLihatFilm();
void menuSewaFilm();
void menuKembalikanFilm();
void menuRiwayatSewa(string nama_pengguna, string password_pengguna);
void menuTambahFilm();
void menuEditFilm();
void menuHapusFilm();
void menuSemuaRiwayat();
void prosesRegister(string role_default);
void inputTanggal(int &hari, int &bulan, int &tahun);

bool prosesLogin(string &nama_hasil, string &password_hasil, string
&role_hasil);
int  hitungTotalBayar(int harga_per_hari, int durasi_hari);
int  hitungDendaRekursif(int selisih_hari);
int  hitungDenda(int selisih_hari);
int  hitungDenda(int harga_per_hari, int selisih_hari);
int  cariFilm(string kode_film);
int  cariSewa(int id_sewa, string password_penyewa);
int  hitungSelisihHari(Tanggal tanggal_wajib, Tanggal tanggal_aktual);

void tampilDetailFilm(Film *penunjuk_film);
void simpanDataSewa(DataSewa *penunjuk_sewa, int id_baru, string kode_film,

```

```

        string judul_film, int harga_hari, int durasi,
        Tanggal tgl_sewa, Tanggal tgl_kembali, int total);

void sortingJudul();
void sortingHarga();
void sortingTahun();

int binarySearchHarga(Film *array_film, int jumlah, int harga_dicari);
void linearSearchKata(Film *array_film, int jumlah, string genre_dicari);

```

3.3 Fitur Inisialisasi Data

```

void inisialisasiData() {
    daftar_user[0] = {"Admin", "000", "Admin"};
    daftar_user[1] = {"Nurul", "2509106033", "User"};
    jumlah_user = 2;

    daftar_film[0] = {"F001", "The Nun II", "Horror", 2023,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[1] = {"F002", "Hereditary", "Horror", 2018,
12000, {"A24", "Amerika"}, true};
    daftar_film[2] = {"F003", "It (2017)", "Horror", 2017,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[3] = {"F004", "Talk to Me", "Horror", 2023,
12000, {"A24", "Amerika"}, true};
    daftar_film[4] = {"F005", "Your Name", "Romance", 2016,
10000, {"Toho", "Jepang"}, true};
    daftar_film[5] = {"F006", "To All The Boys", "Romance", 2018,
10000, {"Netflix", "Amerika"}, true};
    daftar_film[6] = {"F007", "Five Feet Apart", "Romance", 2019,
10000, {"Warner Bros", "Amerika"}, true};
    daftar_film[7] = {"F008", "A Silent Voice", "Romance", 2016,
10000, {"Toho", "Jepang"}, true};
    daftar_film[8] = {"F009", "The Dark Knight", "Action", 2008,
13000, {"Warner Bros", "Amerika"}, true};
    daftar_film[9] = {"F010", "John Wick", "Action", 2014,
13000, {"Lionsgate", "Amerika"}, true};
    daftar_film[10] = {"F011", "Mad Max: Fury Road", "Action", 2015,
13000, {"Warner Bros", "Amerika"}, true};
    daftar_film[11] = {"F012", "Demon Slayer: Mugen", "Action", 2020,
13000, {"Toho", "Jepang"}, true};
    daftar_film[12] = {"F013", "Saltburn", "Thriller", 2023,
12000, {"A24", "Amerika"}, true};
    daftar_film[13] = {"F014", "Gone Girl", "Thriller", 2014,
12000, {"Warner Bros", "Amerika"}, true};
    daftar_film[14] = {"F015", "Bird Box", "Thriller", 2018,
12000, {"Netflix", "Amerika"}, true};

```



```

    daftar_film[15] = {"F016", "Midsommar", "Thriller", 2019,
12000, {"A24", "Amerika"}, true};
    daftar_film[16] = {"F017", "Super Mario Bros Movie", "Comedy", 2023,
10000, {"Nintendo", "Amerika"}, true};
    daftar_film[17] = {"F018", "Puss in Boots 2", "Comedy", 2022,
10000, {"DreamWorks", "Amerika"}, true};
    daftar_film[18] = {"F019", "How To Train Your Dragon", "Animation", 2014,
10000, {"DreamWorks", "Amerika"}, true};
    daftar_film[19] = {"F020", "Spirited Away", "Animation", 2001,
10000, {"Toho", "Jepang"}, true};
    jumlah_film = 20;
}

```

3.4 Fitur Login

```

bool prosesLogin(string &nama_hasil, string &password_hasil, string &role_hasil)
{
    int sisa_percobaan = 3;
    bool login_berhasil = false;

    cin.ignore();
    while (sisa_percobaan > 0 && !login_berhasil) {
        string input_nama, input_password;
        cout << " Nama : "; getline(cin, input_nama);
        cout << " Password : "; getline(cin, input_password);
        for (int i = 0; i < jumlah_user; i++) {
            if (daftar_user[i].nama_pengguna == input_nama &&
                daftar_user[i].password_pengguna == input_password) {
                nama_hasil = daftar_user[i].nama_pengguna;
                password_hasil = daftar_user[i].password_pengguna;
                role_hasil = daftar_user[i].role_pengguna;
                login_berhasil = true;
                break;
            }
        }
        if (login_berhasil) {
            cout << " [v] Selamat datang, " << nama_hasil << "!" << endl;
            cout << " [Tekan Enter...]" << cin.get();
        } else {
            sisa_percobaan--;
            if (sisa_percobaan > 0)
                cout << " [!] Nama/password salah. Sisa percobaan: " <<
sisa_percobaan << endl;
            else
                cout << " [!!!] Gagal 3 kali. Kembali ke menu utama." << endl;
        }
    }
    return login_berhasil;
}

```

```
}
```

3.5 Fitur Register

```
void prosesRegister(string role_default) {
    if (jumlah_user >= MAKS_USER) {
        cout << "    [!] Kapasitas pengguna penuh." << endl;
        cout << "    [Tekan Enter...]" << cin.ignore(); cin.get(); return;
    }
    cin.ignore();
    cout << "    Nama      : "; getline(cin,
daftar_user[jumlah_user].nama_pengguna);
    bool nama_sudah_ada = false;
    for (int i = 0; i < jumlah_user; i++)
        if (daftar_user[i].nama_pengguna ==
daftar_user[jumlah_user].nama_pengguna) {
            nama_sudah_ada = true; break;
        }
    if (nama_sudah_ada) {
        cout << "    [!] Nama sudah digunakan." << endl;
        daftar_user[jumlah_user].nama_pengguna = "";
    } else {
        cout << "    Password  : "; getline(cin,
daftar_user[jumlah_user].password_pengguna);
        daftar_user[jumlah_user].role_pengguna = role_default;
        jumlah_user++;
        cout << "    [v] Akun berhasil dibuat!" << endl;
    }
    cout << "    [Tekan Enter...]" << cin.get();
}
```

3.6 Fitur Lihat Film & Tampil Detail Film

```
void menuLihatFilm() {
    cout << "    Urutkan berdasarkan:" << endl;
    cout << "    1. Judul A-Z          (Bubble Sort)" << endl;
    cout << "    2. Harga Tertinggi (Selection Sort)" << endl;
    cout << "    3. Tahun Terlama    (Insertion Sort)" << endl;
    cout << "    Pilihan : ";
    int pilihan_sort; cin >> pilihan_sort;

    if (pilihan_sort == 1) sortingJudul();
    else if (pilihan_sort == 2) sortingHarga();
    else if (pilihan_sort == 3) sortingTahun();
    else {
        cout << "    [!] Pilihan tidak valid, masukkan 1-3." << endl;
    }
}
```



```

        for (int i = 0; i < jumlah_film; i++) salinan_film[i] = daftar_film[i];

        for (int i = 1; i < jumlah_film; i++) {
            Film film_kunci = salinan_film[i];
            int j = i - 1;
            while (j >= 0 && salinan_film[j].harga_sewa > film_kunci.harga_sewa)
            {
                salinan_film[j + 1] = salinan_film[j];
                j--;
            }
            salinan_film[j + 1] = film_kunci;
        }

        int harga_dicari;
        cout << "    Masukkan harga yang ingin dicari (Rp) : ";
        cin >> harga_dicari;

        int indeks_hasil = binarySearchHarga(salinan_film, jumlah_film,
harga_dicari);

        if (indeks_hasil != -1) {
            cout << "    [v] Film dengan harga Rp" << harga_dicari << "
ditemukan!" << endl;
            tampilDetailFilm(salinan_film + indeks_hasil);

            bool ada_lain = false;
            int kiri = indeks_hasil - 1;
            int kanan = indeks_hasil + 1;
            while (kiri >= 0 && (salinan_film + kiri)->harga_sewa ==
harga_dicari) {
                if (!ada_lain) { cout << "    Film lain dengan harga sama:" <<
endl; ada_lain = true; }
                tampilDetailFilm(salinan_film + kiri);
                kiri--;
            }
            while (kanan < jumlah_film && (salinan_film + kanan)->harga_sewa ==
harga_dicari) {
                if (!ada_lain) { cout << "    Film lain dengan harga sama:" <<
endl; ada_lain = true; }
                tampilDetailFilm(salinan_film + kanan);
                kanan++;
            }
        } else {
            cout << "    [!] Film dengan harga Rp" << harga_dicari << " tidak
ditemukan." << endl;
        }
        cout << "    [Tekan Enter...]"; cin.ignore(); cin.get();

    } else if (pilihan_aksi == 3) {
        string genre_dicari;
        cout << "    Genre yang tersedia: Horror, Romance, Action, Thriller,

```

```

Comedy, Animation" << endl;
    cout << "    Masukkan genre yang dicari : ";
    cin.ignore(); getline(cin, genre_dicari);

    if (genre_dicari != "") {
        cout << "    Hasil pencarian genre \"" << genre_dicari << "\" : " <<
endl;
        linearSearchKata(daftar_film, jumlah_film, genre_dicari);
    } else {
        cout << "    [!] Genre tidak boleh kosong." << endl;
    }
    cout << "    [Tekan Enter...]" << cin.get();
}
}

```

3.7 Fungsi Sorting

```

void sortingJudul() {
    Film film_tukar;
    bool sudah_urut;
    for (int i = 0; i < jumlah_film - 1; i++) {
        sudah_urut = false;
        for (int j = 0; j < jumlah_film - i - 1; j++) {
            if (daftar_film[j].judul_film > daftar_film[j +
1].judul_film) {
                film_tukar          = daftar_film[j];
                daftar_film[j]       = daftar_film[j + 1];
                daftar_film[j + 1]   = film_tukar;
                sudah_urut           = true;
            }
        }
        if (!sudah_urut) break;
    }
}

void sortingHarga() {
    int indeks_terbesar;
    Film film_tukar;
    for (int i = 0; i < jumlah_film - 1; i++) {
        indeks_terbesar = i;
        for (int j = i + 1; j < jumlah_film; j++) {
            if (daftar_film[j].harga_sewa >
daftar_film[indeks_terbesar].harga_sewa) {
                indeks_terbesar = j;
            }
        }
    }
}

```

```

        if (indeks_terbesar != i) {
            film_tukar = daftar_film[i];
            daftar_film[i] = daftar_film[indeks_terbesar];
            daftar_film[indeks_terbesar] = film_tukar;
        }
    }
}

void sortingTahun() {
    Film film_kunci;
    int j;
    for (int i = 1; i < jumlah_film; i++) {
        film_kunci = daftar_film[i];
        j = i - 1;
        while (j >= 0 && daftar_film[j].tahun_rilis >
film_kunci.tahun_rilis) {
            daftar_film[j + 1] = daftar_film[j];
            j = j - 1;
        }
        daftar_film[j + 1] = film_kunci;
    }
}

```

3.8 Fungsi Searching

```

int binarySearchHarga(Film *array_film, int jumlah, int
harga_dicari) {
    int batas_kiri = 0;
    int batas_kanan = jumlah - 1;

    while (batas_kiri <= batas_kanan) {
        int posisi_tengah = batas_kiri + (batas_kanan -
batas_kiri) / 2;

        if ((array_film + posisi_tengah)->harga_sewa ==
harga_dicari) {
            return posisi_tengah;
        } else if ((array_film + posisi_tengah)->harga_sewa <
harga_dicari) {
            batas_kiri = posisi_tengah + 1;
        } else {
            batas_kanan = posisi_tengah - 1;
        }
    }
}

```

```

    }

    return -1;
}

void linearSearchKata(Film *array_film, int jumlah, string
genre_dicari) {
    bool ditemukan = false;

    cout << "
+-----+-----+-----+-----+-----+-----+
----+-----+" << endl;
    cout << " | Kode | Judul | Genre
| Tahun | Harga/Hr | Status |" << endl;
    cout << "
+-----+-----+-----+-----+-----+-----+
----+-----+" << endl;

    for (int i = 0; i < jumlah; i++) {
        Film *penunjuk = array_film + i;

        if (penunjuk->genre_film == genre_dicari) {
            string status_film = penunjuk->tersedia ? "Tersedia" :
"Disewa ";
            string harga_str = "Rp" +
to_string(penunjuk->harga_sewa);
            cout << " | " << left << setw(5) <<
penunjuk->kode_film
                << " | " << setw(29) <<
penunjuk->judul_film
                << " | " << setw(10) <<
penunjuk->genre_film
                << " | " << setw(5) <<
penunjuk->tahun_rilis
                << " | " << setw(8) << harga_str
                << " | " << status_film << " | " << endl;
            ditemukan = true;
        }
    }

    cout << "
+-----+-----+-----+-----+-----+-----+

```

```

----+-----+" << endl;

    if (!ditemukan) {
        cout << "    [!] Film dengan genre \"<
\" tidak ditemukan.\" << endl;
    }
}

```

3.9 Fitur Sewa Film

```

void inputTanggal(int &hari, int &bulan, int &tahun) {
    do {
        cout << "    Hari (1-31) : "; cin >> hari;
        if (hari < 1 || hari > 31) cout << "    [!] Hari tidak valid, masukkan
1-31.\" << endl;
    } while (hari < 1 || hari > 31);

    do {
        cout << "    Bulan (1-12) : "; cin >> bulan;
        if (bulan < 1 || bulan > 12) cout << "    [!] Bulan tidak valid, masukkan
1-12.\" << endl;
    } while (bulan < 1 || bulan > 12);

    do {
        cout << "    Tahun      : "; cin >> tahun;
        if (tahun < 2000) cout << "    [!] Tahun tidak valid, masukkan >= 2000.\"
<< endl;
    } while (tahun < 2000);
}

int hitungTotalBayar(int harga_per_hari, int durasi_hari) {
    return harga_per_hari * durasi_hari;
}

int cariFilm(string kode_film) {
    for (int i = 0; i < jumlah_film; i++) {
        if (daftar_film[i].kode_film == kode_film) return i;
    }
    return -1;
}

void simpanDataSewa(DataSewa *penunjuk_sewa, int id_baru, string kode_film,
                    string judul_film, int harga_hari, int durasi,
                    Tanggal tgl_sewa, Tanggal tgl_kembali, int total) {
    (*penunjuk_sewa).id_sewa      = id_baru;

```



```

(*penunjuk_sewa).password_penyewa = password_aktif;
(*penunjuk_sewa).nama_penyewa      = nama_aktif;
(*penunjuk_sewa).kode_film_sewa     = kode_film;
(*penunjuk_sewa).judul_film_sewa    = judul_film;
(*penunjuk_sewa).harga_per_hari     = harga_hari;
(*penunjuk_sewa).durasi_hari         = durasi;
(*penunjuk_sewa).tanggal_sewa       = tgl_sewa;
(*penunjuk_sewa).tanggal_kembali    = tgl_kembali;
(*penunjuk_sewa).total_bayar        = total;
(*penunjuk_sewa).denda               = 0;
(*penunjuk_sewa).sudah_kembali      = false;
}

void menuSewaFilm() {
    cout << "   === SEWA FILM ===" << endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;
    cout << "| Kode   | Judul                               | Genre   | Harga/Hr |" <<
endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;

    bool ada_tersedia = false;
    for (int i = 0; i < jumlah_film; i++) {
        if (!daftar_film[i].tersedia) continue;
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "| " << left << setw(5) << daftar_film[i].kode_film
            << "| " << setw(29) << daftar_film[i].judul_film
            << "| " << setw(10) << daftar_film[i].genre_film
            << "| " << setw(8) << harga_str << "|" << endl;
        ada_tersedia = true;
    }
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;

    if (!ada_tersedia) {
        cout << " Semua film sedang disewa." << endl;
        cin.ignore(); cin.get(); return;
    }

    string kode_dipilih;
    cout << " Kode film (Enter -> Kembali) : ";
    cin.ignore(); getline(cin, kode_dipilih);
    if (kode_dipilih == "") return;

    int indeks_film = cariFilm(kode_dipilih);
    if (indeks_film == -1 || !daftar_film[indeks_film].tersedia) {
        cout << "  [] Film tidak ditemukan atau sedang disewa." << endl;
        cin.get(); return;
    }
}

```

```

    Tanggal tanggal_sewa;
    int durasi_hari;
    cout << "    Tanggal Sewa:" << endl;
    inputTanggal(tanggal_sewa.hari, tanggal_sewa.bulan, tanggal_sewa.tahun);
    do {
        cout << "    Durasi (hari, min 1) : "; cin >> durasi_hari;
        if (durasi_hari < 1) cout << "    [!] Durasi tidak valid, minimal 1 hari."
<< endl;
    } while (durasi_hari < 1);

    Tanggal tanggal_kembali = tanggal_sewa;
    int hari_per_bulan[13] = {0, 31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30,
31};
    tanggal_kembali.hari += durasi_hari;
    while (tanggal_kembali.hari > hari_per_bulan[tanggal_kembali.bulan]) {
        tanggal_kembali.hari -= hari_per_bulan[tanggal_kembali.bulan];
        tanggal_kembali.bulan++;
        if (tanggal_kembali.bulan > 12) { tanggal_kembali.bulan = 1;
tanggal_kembali.tahun++; }
    }

    int total_bayar_sewa = hitungTotalBayar(daftar_film[indeks_film].harga_sewa,
durasi_hari);

    cout << "    Film          : " << daftar_film[indeks_film].judul_film << endl;
    cout << "    Tgl Sewa       : ";
    if (tanggal_sewa.hari < 10) cout << "0"; cout << tanggal_sewa.hari <<
"/";
    if (tanggal_sewa.bulan < 10) cout << "0"; cout << tanggal_sewa.bulan <<
"/" << tanggal_sewa.tahun << endl;
    cout << "    Wajib Kembali : ";
    if (tanggal_kembali.hari < 10) cout << "0"; cout << tanggal_kembali.hari
<< "/";
    if (tanggal_kembali.bulan < 10) cout << "0"; cout << tanggal_kembali.bulan
<< "/" << tanggal_kembali.tahun << endl;
    cout << "    Total Bayar   : Rp" << total_bayar_sewa << endl;
    cout << "    Denda         : Rp" << DENDA_PER_HARI << "/hari jika terlambat"
<< endl;
    cout << "    Konfirmasi? (y/n) : ";
    char konfirmasi_sewa; cin >> konfirmasi_sewa;
    if (konfirmasi_sewa != 'y' && konfirmasi_sewa != 'Y') {
        cout << "    Dibatalkan." << endl; cin.ignore(); cin.get(); return;
    }

    daftar_film[indeks_film].tersedia = false;

    DataSewa *penunjuk_sewa = &daftar_sewa[jumlah_sewa];
    simpanDataSewa(penunjuk_sewa, id_berikutnya++, kode_dipilih,
        daftar_film[indeks_film].judul_film,
        daftar_film[indeks_film].harga_sewa,
        durasi_hari, tanggal_sewa, tanggal_kembali,

```

```

total_bayar_sewa);
    jumlah_sewa++;

    cout << "    [v] Berhasil menyewa \"" << daftar_film[indeks_film].judul_film
<< "\"!" << endl;
    cout << "    [Tekan Enter...]">
    cin.ignore(); cin.get();
}

```

3.10 Fitur Kembalikan Film

```

int hitungSelisihHari(Tanggal tanggal_wajib, Tanggal tanggal_aktual) {
    int total_wajib = tanggal_wajib.tahun * 365 + tanggal_wajib.bulan * 30 +
tanggal_wajib.hari;
    int total_aktual = tanggal_aktual.tahun * 365 + tanggal_aktual.bulan * 30 +
tanggal_aktual.hari;
    return total_aktual - total_wajib;
}

int hitungDendaRekursif(int selisih_hari) {
    if (selisih_hari <= 0) return 0;
    return DENDA_PER_HARI + hitungDendaRekursif(selisih_hari - 1);
}

int hitungDenda(int selisih_hari) {
    if (selisih_hari <= 0) return 0;
    return hitungDendaRekursif(selisih_hari);
}

int hitungDenda(int harga_per_hari, int selisih_hari) {
    if (selisih_hari <= 0) return 0;
    return selisih_hari * harga_per_hari;
}

int cariSewa(int id_sewa, string password_penyewa) {
    for (int i = 0; i < jumlah_sewa; i++) {
        if (daftar_sewa[i].id_sewa == id_sewa &&
            daftar_sewa[i].password_penyewa == password_penyewa &&
            !daftar_sewa[i].sudah_kembali) return i;
    }
    return -1;
}

void menuKembalikanFilm() {
    cout << "    === KEMBALIKAN FILM ===" << endl;
    cout << "+-----+-----+-----+-----+-----+-----+" <<
endl;
    cout << "| ID | Judul                               | Tgl Sewa    | Wajib Kemb |" <<
endl;

```

```

        cout << "+----+-----+-----+-----+" <<
endl;

        bool ada_sewa_aktif = false;
        for (int i = 0; i < jumlah_sewa; i++) {
            if (daftar_sewa[i].password_penyewa != password_aktif ||
daftar_sewa[i].sudah_kembali) continue;

            string tgl_sewa_str = "";
            if (daftar_sewa[i].tanggal_sewa.hari < 10) tgl_sewa_str += "0";
            tgl_sewa_str += to_string(daftar_sewa[i].tanggal_sewa.hari) + "/";
            if (daftar_sewa[i].tanggal_sewa.bulan < 10) tgl_sewa_str += "0";
            tgl_sewa_str += to_string(daftar_sewa[i].tanggal_sewa.bulan) + "/" +
to_string(daftar_sewa[i].tanggal_sewa.tahun);

            string tgl_kemb_str = "";
            if (daftar_sewa[i].tanggal_kembali.hari < 10) tgl_kemb_str += "0";
            tgl_kemb_str += to_string(daftar_sewa[i].tanggal_kembali.hari) + "/";
            if (daftar_sewa[i].tanggal_kembali.bulan < 10) tgl_kemb_str += "0";
            tgl_kemb_str += to_string(daftar_sewa[i].tanggal_kembali.bulan) + "/" +
to_string(daftar_sewa[i].tanggal_kembali.tahun);

            cout << "| " << left << setw(2) << daftar_sewa[i].id_sewa
                << " | " << setw(29) << daftar_sewa[i].judul_film_sewa
                << " | " << setw(11) << tgl_sewa_str
                << " | " << setw(11) << tgl_kemb_str << "|" << endl;
            ada_sewa_aktif = true;
        }
        cout << "+----+-----+-----+-----+" <<
endl;

        if (!ada_sewa_aktif) {
            cout << " Tidak ada sewa aktif." << endl; cin.ignore(); cin.get();
return;
        }

        int id_dikembalikan;
        cout << " ID sewa : "; cin >> id_dikembalikan;

        int indeks_sewa = cariSewa(id_dikembalikan, password_aktif);
        if (indeks_sewa == -1) {
            cout << " [!] ID tidak valid." << endl; cin.ignore(); cin.get();
return;
        }

        Tanggal tanggal_aktual_kembali;
        cout << " Tanggal Kembali:" << endl;
        inputTanggal(tanggal_aktual_kembali.hari, tanggal_aktual_kembali.bulan,
tanggal_aktual_kembali.tahun);

        int selisih_hari =

```

```

hitungSelisihHari(daftar_sewa[indeks_sewa].tanggal_kembali,
tanggal_aktual_kembali);
    int denda_terlambat = hitungDenda(selisih_hari);

    daftar_sewa[indeks_sewa].denda = denda_terlambat;
    daftar_sewa[indeks_sewa].sudah_kembali = true;

    int indeks_film_kembali = cariFilm(daftar_sewa[indeks_sewa].kode_film_sewa);
    if (indeks_film_kembali != -1) daftar_film[indeks_film_kembali].tersedia =
true;

    cout << " [v] Film berhasil dikembalikan!" << endl;
    if (selisih_hari > 0) {
        cout << " Terlambat " << selisih_hari << " hari" << endl;
        cout << " Denda : Rp" << denda_terlambat << endl;
    } else {
        cout << " Tepat waktu, tidak ada denda." << endl;
    }
    cout << " TOTAL BAYAR: Rp" << daftar_sewa[indeks_sewa].total_bayar +
denda_terlambat << endl;
    cout << " [Tekan Enter...]"; cin.ignore(); cin.get();
}

```

3.11 Fitur Riwayat Sewa

```

void menuRiwayatSewa(string nama_pengguna, string password_pengguna) {
    cout << " === RIWAYAT SEWA - " << nama_pengguna << " ===" << endl;
    cout << "+---+-----+-----+-----+"
<< endl;
    cout << "| ID | Judul | Total | Status |"
<< endl;
    cout << "+---+-----+-----+-----+"
<< endl;

    bool ada_riwayat = false;
    for (int i = 0; i < jumlah_sewa; i++) {
        if (daftar_sewa[i].password_penyewa != password_pengguna)
continue;
        string status_sewa = daftar_sewa[i].sudah_kembali ? "Selesai "
: "Aktif ";
        int total_tampil = daftar_sewa[i].total_bayar +
daftar_sewa[i].denda;
        string total_str = "Rp" + to_string(total_tampil);
        cout << "| " << left << setw(2) << daftar_sewa[i].id_sewa
<< " | " << setw(29) <<

```

```

daftar_sewa[i].judul_film_sewa
    << " | "          << setw(8) << total_str
    << " | "          << status_sewa << " | " << endl;
    ada_riwayat = true;
}
cout << "+-----+-----+-----+-----+-----+"
<< endl;
if (!ada_riwayat) cout << "  Belum ada riwayat." << endl;
cout << "  [Tekan Enter...]" << cin.ignore(); cin.get();
}

```

3.12 Fitur Tambah Film (Admin)

```

void menuTambahFilm() {
    cout << "  === TAMBAH FILM ===" << endl;
    if (jumlah_film >= MAKS_FILM) {
        cout << "  [!] Kapasitas penuh." << endl; cin.ignore(); cin.get();
    }
    return;
}
cout <<
"+-----+-----+-----+-----+-----+" <<
endl;
cout << " | Kode   | Judul                               | Genre   | Tahun   |
Harga/Hr |" << endl;
cout <<
"+-----+-----+-----+-----+-----+" <<
endl;
for (int i = 0; i < jumlah_film; i++) {
    string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
    cout << " | " << left << setw(5) << daftar_film[i].kode_film
    << " | "          << setw(29) << daftar_film[i].judul_film
    << " | "          << setw(10) << daftar_film[i].genre_film
    << " | "          << setw(5)  << daftar_film[i].tahun_rilis
    << " | "          << setw(8)  << harga_str << " | " << endl;
}
cout <<
"+-----+-----+-----+-----+-----+" <<
endl;
cin.ignore();
cout << "  Kode   : "; getline(cin, daftar_film[jumlah_film].kode_film);
if (daftar_film[jumlah_film].kode_film == "") return;

bool kode_duplikat = false;
for (int i = 0; i < jumlah_film; i++)
    if (daftar_film[i].kode_film == daftar_film[jumlah_film].kode_film) {

```

```

kode_duplikat = true; break; }
    if (kode_duplikat) {
        cout << "    [!] Kode sudah ada." << endl;
        daftar_film[jumlah_film].kode_film = ""; cin.get(); return;
    }
    cout << "    Judul   : "; getline(cin, daftar_film[jumlah_film].judul_film);
    cout << "    Genre   : "; getline(cin, daftar_film[jumlah_film].genre_film);
    cout << "    Tahun   : "; cin >> daftar_film[jumlah_film].tahun_rilis;
cin.ignore();
    cout << "    Harga   : "; cin >> daftar_film[jumlah_film].harga_sewa;
cin.ignore();
    cout << "    Studio  : "; getline(cin,
daftar_film[jumlah_film].studio.nama_studio);
    cout << "    Negara  : "; getline(cin,
daftar_film[jumlah_film].studio.negara_studio);
    daftar_film[jumlah_film].tersedia = true;
    jumlah_film++;
    cout << "    [v] Film berhasil ditambahkan!" << endl;
    cout << "    [Tekan Enter...]"; cin.get();
}

```

3.13 Fitur Edit Film (Admin)

```

void menuEditFilm() {
    cout << "    === EDIT FILM ===" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
    endl;
    cout << "    | Kode   | Judul                               | Genre   | Tahun   |
    Harga/Hr |" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
    endl;
    for (int i = 0; i < jumlah_film; i++) {
        string harga_str = "Rp" + to_string(daftar_film[i].harga_sewa);
        cout << "    | " << left << setw(5) << daftar_film[i].kode_film
            << " | " << setw(29) << daftar_film[i].judul_film
            << " | " << setw(10) << daftar_film[i].genre_film
            << " | " << setw(5) << daftar_film[i].tahun_rilis
            << " | " << setw(8) << harga_str << " |" << endl;
    }
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
    endl;
}

```

```

    cin.ignore();
    string kode_diedit;
    cout << "   Kode film (Enter -> Kembali) : ";
    getline(cin, kode_diedit);
    if (kode_diedit == "") return;

    int indeks_edit = cariFilm(kode_diedit);
    if (indeks_edit == -1) {
        cout << "   [!] Kode tidak ditemukan." << endl; cin.get(); return;
    }
    cout << "   [1]Judul   [2]Genre   [3]Tahun   [4]Harga   [5]Studio   [6]Negara" <<
endl;
    cout << "   Pilihan   : "; int pilihan_edit; cin >> pilihan_edit;
cin.ignore();
    if (pilihan_edit == 1) { cout << "   Judul baru   : "; getline(cin,
daftar_film[indeks_edit].judul_film); }
    else if (pilihan_edit == 2) { cout << "   Genre baru   : "; getline(cin,
daftar_film[indeks_edit].genre_film); }
    else if (pilihan_edit == 3) { cout << "   Tahun baru   : "; cin >>
daftar_film[indeks_edit].tahun_rilis; }
    else if (pilihan_edit == 4) { cout << "   Harga baru   : "; cin >>
daftar_film[indeks_edit].harga_sewa; }
    else if (pilihan_edit == 5) { cout << "   Studio baru : "; getline(cin,
daftar_film[indeks_edit].studio.nama_studio); }
    else if (pilihan_edit == 6) { cout << "   Negara baru : "; getline(cin,
daftar_film[indeks_edit].studio.negara_studio); }
    cout << "   [v] Film berhasil diedit!" << endl;
    cout << "   [Tekan Enter...]" << cin.get();
}

```

3.14 Fitur Hapus Film (Admin)

```

void menuHapusFilm() {
    cout << "   === HAPUS FILM ===" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
endl;
    cout << " | Kode   | Judul                               | Genre   | Tahun | Status
| " << endl;
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
endl;
    for (int i = 0; i < jumlah_film; i++) {
        string status_film = daftar_film[i].tersedia ? "Tersedia" : "Disewa ";
        cout << " | " << left << setw(5) << daftar_film[i].kode_film
        << " | "
        << setw(29) << daftar_film[i].judul_film
        << " | "
        << setw(10) << daftar_film[i].genre_film

```



```

        << " | "          << setw(5) << daftar_film[i].tahun_rilis
        << " | "          << status_film << " |" << endl;
    }
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
    endl;
    cin.ignore();
    string kode_dihapus;
    cout << "   Kode film (Enter -> Kembali) : ";
    getline(cin, kode_dihapus);
    if (kode_dihapus == "") return;

    int indeks_hapus = cariFilm(kode_dihapus);
    if (indeks_hapus == -1) {
        cout << "   [!] Kode tidak ditemukan." << endl; cin.get(); return;
    }
    if (!daftar_film[indeks_hapus].tersedia) {
        cout << "   [!] Film sedang disewa, tidak bisa dihapus." << endl;
    }
    cin.get(); return;
}
    cout << "   Hapus \"" << daftar_film[indeks_hapus].judul_film << "\"? (y/n) :
";
    char konfirmasi_hapus; cin >> konfirmasi_hapus;
    if (konfirmasi_hapus != 'y' && konfirmasi_hapus != 'Y') {
        cout << "   Dibatalkan." << endl; cin.ignore(); cin.get(); return;
    }
    for (int i = indeks_hapus; i < jumlah_film - 1; i++) daftar_film[i] =
daftar_film[i + 1];
    jumlah_film--;
    cout << "   [v] Film berhasil dihapus!" << endl;
    cout << "   [Tekan Enter...]"; cin.ignore(); cin.get();
}

```

3.15 Fitur Semua Riwayat (Admin)

```

void menuSemuaRiwayat() {
    cout << "   === SEMUA RIWAYAT SEWA ===" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
    endl;
    cout << " | ID | Penyewa      | Film                      | Total      | Status
 |" << endl;
    cout <<
    "+-----+-----+-----+-----+-----+-----+" <<
    endl;
    if (jumlah_sewa == 0) cout << "   Belum ada riwayat." << endl;
    for (int i = 0; i < jumlah_sewa; i++) {
        string status_sewa = daftar_sewa[i].sudah_kembali ? "Selesai " : "Aktif

```



```

        cout << "    2. Sewa Film"    << endl;
        cout << "    3. Kembalikan" << endl;
        cout << "    4. Riwayat Sewa" << endl;
        if (role_aktif == "Admin") {
            cout << "-----" <<
endl;
            cout << "    5. Tambah Film"    << endl;
            cout << "    6. Edit Film"    << endl;
            cout << "    7. Hapus Film"    << endl;
            cout << "    8. Semua Riwayat" << endl;
        }
        cout << "    0. Logout" << endl;
        cout << "-----" <<
endl;

        cout << "    Pilihan : ";
        int pilihan_menu; cin >> pilihan_menu;

        switch (pilihan_menu) {
            case 1: menuLihatFilm();
break;

            case 2: menuSewaFilm();
break;

            case 3: menuKembalikanFilm();
break;

            case 4: menuRiwayatSewa(nama_aktif, password_aktif);
break;

            case 5:
                if (role_aktif == "Admin") menuTambahFilm();
                else { cout << "    [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                break;
            case 6:
                if (role_aktif == "Admin") menuEditFilm();
                else { cout << "    [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                break;
            case 7:
                if (role_aktif == "Admin") menuHapusFilm();
                else { cout << "    [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                break;
            case 8:
                if (role_aktif == "Admin") menuSemuaRiwayat();
                else { cout << "    [!] Akses ditolak." << endl;
cin.ignore(); cin.get(); }
                break;
            case 0:
                cout << "    Sampai jumpa, " << nama_aktif << "!" <<
endl;
                nama_aktif = ""; password_aktif = ""; role_aktif =
"";

```

```

        sedang_login = false;
        cout << " [Tekan Enter...]">> cin.ignore();
cin.get();

        break;
    default:
        cout << " [!] Pilihan tidak valid." << endl;
        cout << " [Tekan Enter...]">> cin.ignore();
cin.get();

        break;
    }
} while (sedang_login);
}
} else if (pilihan_awal == 0) {
    cout << " Terima kasih! Sampai jumpa!" << endl;
    program_berjalan = false;
} else {
    cout << " [!] Pilihan tidak valid." << endl;
    cout << " [Tekan Enter...]">> cin.ignore(); cin.get();
}
} while (program_berjalan);
return 0;
}

```

4. Hasil Output

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : █
```

Gambar 4.1 Menu Login

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : 1
Nama      : Nana
Password  : 009
[!] Salah. Sisa percobaan: 2

Nama      : Nala
Password  : 008
[!] Salah. Sisa percobaan: 1

Nama      : Nurul
Password  : 2509106044
[!!!] Gagal 3 kali. Program berhenti.
```

Gambar 4.2 Login Gagal

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : 1
Nama      : Nurul
Password  : 2509106033
[v] Selamat datang, Nurul!
[Tekan Enter...] █
```

Gambar 4.3 Login Berhasil

```
=====
                SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====
Pilihan : 0
Terima kasih! Sampai jumpa!
PS C:\Users\LOQ\Documents\praktikum-apl\post-test\post-test-apl-2> █
```

Gambar 4.4 Keluar

```
=====
                        SELAMAT DATANG DI CINEMARENT
1. Login
2. Register
0. Keluar
=====

Pilihan : 2
Nama      : Nana
Password  : 009
[v] Akun berhasil dibuat!
[Tekan Enter...]
```

Gambar 4.5 Register

```
=====
MENU USER | Nurul
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
0. Logout
=====

Pilihan : █
```

Gambar 4.6 Menu User

```
=====
Pilihan : 1
Urutkan berdasarkan:
1. Judul A-Z      (Bubble Sort)
2. Harga Tertinggi (Selection Sort)
3. Tahun Terlama  (Insertion Sort)
Pilihan : █
```

Gambar 4.7 Menu User Pilihan 1

```
Pilihan : 1
+-----+-----+-----+-----+-----+-----+
| Kode | Judul                | Genre  | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+-----+
| F008 | A Silent Voice        | Romance | 2016  | Rp10000  | Tersedia |
| F015 | Bird Box              | Thriller | 2018  | Rp12000  | Tersedia |
| F012 | Demon Slayer: Mugen   | Action  | 2020  | Rp13000  | Tersedia |
| F007 | Five Feet Apart       | Romance  | 2019  | Rp10000  | Tersedia |
| F014 | Gone Girl             | Thriller | 2014  | Rp12000  | Tersedia |
| F002 | Hereditary            | Horror  | 2018  | Rp12000  | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014  | Rp10000  | Tersedia |
| F003 | It (2017)            | Horror  | 2017  | Rp12000  | Tersedia |
| F010 | John Wick             | Action  | 2014  | Rp13000  | Tersedia |
| F011 | Mad Max: Fury Road    | Action  | 2015  | Rp13000  | Tersedia |
| F016 | Midsommar             | Thriller | 2019  | Rp12000  | Tersedia |
| F018 | Puss in Boots 2       | Comedy  | 2022  | Rp10000  | Tersedia |
| F013 | Saltburn              | Thriller | 2023  | Rp12000  | Tersedia |
| F020 | Spirited Away         | Animation | 2001  | Rp10000  | Tersedia |
| F017 | Super Mario Bros Movie | Comedy  | 2023  | Rp10000  | Tersedia |
| F004 | Talk to Me            | Horror  | 2023  | Rp12000  | Tersedia |
| F009 | The Dark Knight       | Action  | 2008  | Rp13000  | Tersedia |
| F001 | The Nun II            | Horror  | 2023  | Rp12000  | Tersedia |
| F006 | To All The Boys       | Romance  | 2018  | Rp10000  | Tersedia |
| F005 | Your Name             | Romance  | 2016  | Rp10000  | Tersedia |
+-----+-----+-----+-----+-----+-----+

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : █
```

Gambar 4.8 User Pilih Urutan 1

Pilihan : 2

Kode	Judul	Genre	Tahun	Harga/Hr	Status
F012	Demon Slayer: Mugen	Action	2020	Rp13000	Tersedia
F010	John Wick	Action	2014	Rp13000	Tersedia
F011	Mad Max: Fury Road	Action	2015	Rp13000	Tersedia
F009	The Dark Knight	Action	2008	Rp13000	Tersedia
F014	Gone Girl	Thriller	2014	Rp12000	Tersedia
F002	Hereditary	Horror	2018	Rp12000	Tersedia
F003	It (2017)	Horror	2017	Rp12000	Tersedia
F015	Bird Box	Thriller	2018	Rp12000	Tersedia
F016	Midsommar	Thriller	2019	Rp12000	Tersedia
F013	Saltburn	Thriller	2023	Rp12000	Tersedia
F004	Talk to Me	Horror	2023	Rp12000	Tersedia
F001	The Nun II	Horror	2023	Rp12000	Tersedia
F008	A Silent Voice	Romance	2016	Rp10000	Tersedia
F020	Spirited Away	Animation	2001	Rp10000	Tersedia
F017	Super Mario Bros Movie	Comedy	2023	Rp10000	Tersedia
F019	How To Train Your Dragon	Animation	2014	Rp10000	Tersedia
F007	Five Feet Apart	Romance	2019	Rp10000	Tersedia
F018	Puss in Boots 2	Comedy	2022	Rp10000	Tersedia
F006	To All The Boys	Romance	2018	Rp10000	Tersedia
F005	Your Name	Romance	2016	Rp10000	Tersedia

Pilih aksi:
 1. Lihat detail film (by kode)
 2. Cari film by harga [Binary Search]
 3. Cari film by genre [Linear Search]
 0. Kembali
 Pilihan : █

Gambar 4.9 User Pilih Urutan 2

Pilihan : 3

Kode	Judul	Genre	Tahun	Harga/Hr	Status
F020	Spirited Away	Animation	2001	Rp10000	Tersedia
F009	The Dark Knight	Action	2008	Rp13000	Tersedia
F010	John Wick	Action	2014	Rp13000	Tersedia
F014	Gone Girl	Thriller	2014	Rp12000	Tersedia
F019	How To Train Your Dragon	Animation	2014	Rp10000	Tersedia
F011	Mad Max: Fury Road	Action	2015	Rp13000	Tersedia
F008	A Silent Voice	Romance	2016	Rp10000	Tersedia
F005	Your Name	Romance	2016	Rp10000	Tersedia
F003	It (2017)	Horror	2017	Rp12000	Tersedia
F002	Hereditary	Horror	2018	Rp12000	Tersedia
F015	Bird Box	Thriller	2018	Rp12000	Tersedia
F006	To All The Boys	Romance	2018	Rp10000	Tersedia
F016	Midsommar	Thriller	2019	Rp12000	Tersedia
F007	Five Feet Apart	Romance	2019	Rp10000	Tersedia
F012	Demon Slayer: Mugen	Action	2020	Rp13000	Tersedia
F018	Puss in Boots 2	Comedy	2022	Rp10000	Tersedia
F013	Saltburn	Thriller	2023	Rp12000	Tersedia
F004	Talk to Me	Horror	2023	Rp12000	Tersedia
F001	The Nun II	Horror	2023	Rp12000	Tersedia
F017	Super Mario Bros Movie	Comedy	2023	Rp10000	Tersedia

Pilih aksi:
 1. Lihat detail film (by kode)
 2. Cari film by harga [Binary Search]
 3. Cari film by genre [Linear Search]
 0. Kembali
 Pilihan : █

Gambar 4.10 User Pilih Urut 3

```

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 1
Kode film : F001
+-----+
|  DETAIL FILM  |
+-----+
Kode       : F001
Judul      : The Nun II
Genre      : Horror
Tahun Rilis : 2023
Harga/Hari : Rp12000
Studio     : Warner Bros
Negara     : Amerika
Status     : Tersedia
+-----+
[Tekan Enter...]

```

Gambar 4.11 Hasil Pilih Aksi 1

```

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 2
Masukkan harga yang ingin dicari (Rp) : 13000
[v] Film dengan harga Rp13000 ditemukan!
+-----+
|  DETAIL FILM  |
+-----+
Kode       : F010
Judul      : John Wick
Genre      : Action
Tahun Rilis : 2014
Harga/Hari : Rp13000
Studio     : Lionsgate
Negara     : Amerika
Status     : Tersedia
+-----+
Film lain dengan harga sama:
+-----+
|  DETAIL FILM  |
+-----+
Kode       : F009
Judul      : The Dark Knight
Genre      : Action
Tahun Rilis : 2008
Harga/Hari : Rp13000
Studio     : Warner Bros
Negara     : Amerika
Status     : Tersedia
+-----+

```

Gambar 4.12 Hasil Pilih Aksi 2


```

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 3
Genre yang tersedia: Horror, Romance, Action, Thriller, Comedy, Animation
Masukkan genre yang dicari : Comedy
Hasil pencarian genre "Comedy" :
+-----+-----+-----+-----+-----+
| Kode | Judul                | Genre  | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+
| F018 | Puss in Boots 2      | Comedy | 2022  | Rp10000  | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023  | Rp10000  | Tersedia |
+-----+-----+-----+-----+-----+
[Tekan Enter...]

```

Gambar 4.13 Hasil Pilih Aksi 3

```

Pilih aksi:
1. Lihat detail film (by kode)
2. Cari film by harga      [Binary Search]
3. Cari film by genre     [Linear Search]
0. Kembali
Pilihan : 0
=====
MENU USER | Nurul
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
0. Logout
-----
Pilihan : 

```

Gambar 4.14 Hasil Pilih Aksi 0

```

Pilihan : 2
=== SEWA FILM ===
+-----+-----+-----+-----+
| Kode | Judul | Genre | Harga/Hr |
+-----+-----+-----+-----+
| F001 | The Nun II | Horror | Rp12000 |
| F002 | Hereditary | Horror | Rp12000 |
| F003 | It (2017) | Horror | Rp12000 |
| F004 | Talk to Me | Horror | Rp12000 |
| F005 | Your Name | Romance | Rp10000 |
| F006 | To All The Boys | Romance | Rp10000 |
| F007 | Five Feet Apart | Romance | Rp10000 |
| F008 | A Silent Voice | Romance | Rp10000 |
| F009 | The Dark Knight | Action | Rp13000 |
| F010 | John Wick | Action | Rp13000 |
| F011 | Mad Max: Fury Road | Action | Rp13000 |
| F012 | Demon Slayer: Mugen | Action | Rp13000 |
| F013 | Saltburn | Thriller | Rp12000 |
| F014 | Gone Girl | Thriller | Rp12000 |
| F015 | Bird Box | Thriller | Rp12000 |
| F016 | Midsommar | Thriller | Rp12000 |
| F017 | Super Mario Bros Movie | Comedy | Rp10000 |
| F018 | Puss in Boots 2 | Comedy | Rp10000 |
| F019 | How To Train Your Dragon | Animation | Rp10000 |
| F020 | Spirited Away | Animation | Rp10000 |
+-----+-----+-----+-----+

Kode film (Enter -> Kembali) : F012
Tanggal Sewa:
Hari (1-31) : 4
Bulan (1-12) : 6
Tahun : 2025
Durasi (hari, min 1) : 6
Film : Demon Slayer: Mugen
Tgl Sewa : 04/06/2025
Wajib Kembali : 10/06/2025
Total Bayar : Rp78000
Denda : Rp5000/hari jika terlambat
Konfirmasi? (y/n) : y
[v] Berhasil menyewa "Demon Slayer: Mugen"!
[Tekan Enter...]

```

Gambar 4.15 Menu User Pilihan 2

```

Kode film (Enter -> Kembali) : F012
Tanggal Sewa:
Hari (1-31) : 4
Bulan (1-12) : 6
Tahun : 2025
Durasi (hari, min 1) : 6
Film : Demon Slayer: Mugen
Tgl Sewa : 04/06/2025
Wajib Kembali : 10/06/2025
Total Bayar : Rp78000
Denda : Rp5000/hari jika terlambat
Konfirmasi? (y/n) : n
Dibatalkan.

```

Gambar 4.16 Film Tidak Jadi Disewa

```

-----
Pilihan : 3
=== KEMBALIKAN FILM ===
+-----+-----+-----+
| ID | Judul                  | Tgl Sewa  | Wajib Kemb |
+-----+-----+-----+
| 1  | Demon Slayer: Mugen    | 04/06/2025 | 10/06/2025 |
+-----+-----+-----+

ID sewa : 1
Tanggal Kembali:
Hari (1-31) : 9
Bulan (1-12) : 6
Tahun      : 2025
[v] Film berhasil dikembalikan!
Tepat waktu, tidak ada denda.
TOTAL BAYAR: Rp78000
[Tekan Enter...]

```

Gambar 4.17 Menu User Pilihan 3

```

Pilihan : 3
=== KEMBALIKAN FILM ===
+-----+-----+-----+
| ID | Judul                  | Tgl Sewa  | Wajib Kemb |
+-----+-----+-----+
| 1  | Demon Slayer: Mugen    | 04/06/2025 | 10/06/2025 |
+-----+-----+-----+

ID sewa : 1
Tanggal Kembali:
Hari (1-31) : 11
Bulan (1-12) : 6
Tahun      : 2025
[v] Film berhasil dikembalikan!
Terlambat 1 hari
Denda      : Rp5000
TOTAL BAYAR: Rp83000
[Tekan Enter...]

```

Gambar 4.18 Jika Terlambat

```

-----
Pilihan : 4
=== RIWAYAT SEWA - Nurul ===
+-----+-----+-----+
| ID | Judul                  | Total     | Status     |
+-----+-----+-----+
| 1  | Demon Slayer: Mugen    | Rp83000   | Selesai    |
+-----+-----+-----+

[Tekan Enter...]

```

Gambar 4.19 Menu User Pilihan 4

```

=====
MENU ADMIN | Admin
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
-----
5. Tambah Film
6. Edit Film
7. Hapus Film
8. Semua Riwayat
0. Logout
-----
Pilihan : 

```

Gambar 4.20 Menu Admin

Pilihan : 5
 === TAMBAH FILM ===
 Daftar film saat ini:

Kode	Judul	Genre	Tahun	Harga/Hr
F001	The Nun II	Horror	2023	Rp12000
F002	Hereditary	Horror	2018	Rp12000
F003	It (2017)	Horror	2017	Rp12000
F004	Talk to Me	Horror	2023	Rp12000
F005	Your Name	Romance	2016	Rp10000
F006	To All The Boys	Romance	2018	Rp10000
F007	Five Feet Apart	Romance	2019	Rp10000
F008	A Silent Voice	Romance	2016	Rp10000
F009	The Dark Knight	Action	2008	Rp13000
F010	John Wick	Action	2014	Rp13000
F011	Mad Max: Fury Road	Action	2015	Rp13000
F012	Demon Slayer: Mugen	Action	2020	Rp13000
F013	Saltburn	Thriller	2023	Rp12000
F014	Gone Girl	Thriller	2014	Rp12000
F015	Bird Box	Thriller	2018	Rp12000
F016	Midsommar	Thriller	2019	Rp12000
F017	Super Mario Bros Movie	Comedy	2023	Rp10000
F018	Puss in Boots 2	Comedy	2022	Rp10000
F019	How To Train Your Dragon	Animation	2014	Rp10000
F020	Spirited Away	Animation	2001	Rp10000

Kode : F021
 Judul : My Neighbor Totoro
 Genre : Comedy
 Tahun : 1998
 Harga : 12000
 Studio : Ghibli
 Negara : Jepang
 [v] Film ditambahkan!
 [Tekan Enter...]

Gambar 4.21 Menu Admin Pilihan 5

Pilihan : 1

Kode	Judul	Genre	Tahun	Harga/Hr	Status
F001	The Nun II	Horror	2023	Rp12000	Tersedia
F002	Hereditary	Horror	2018	Rp12000	Tersedia
F003	It (2017)	Horror	2017	Rp12000	Tersedia
F004	Talk to Me	Horror	2023	Rp12000	Tersedia
F005	Your Name	Romance	2016	Rp10000	Tersedia
F006	To All The Boys	Romance	2018	Rp10000	Tersedia
F007	Five Feet Apart	Romance	2019	Rp10000	Tersedia
F008	A Silent Voice	Romance	2016	Rp10000	Tersedia
F009	The Dark Knight	Action	2008	Rp13000	Tersedia
F010	John Wick	Action	2014	Rp13000	Tersedia
F011	Mad Max: Fury Road	Action	2015	Rp13000	Tersedia
F012	Demon Slayer: Mugen	Action	2020	Rp13000	Tersedia
F013	Saltburn	Thriller	2023	Rp12000	Tersedia
F014	Gone Girl	Thriller	2014	Rp12000	Tersedia
F015	Bird Box	Thriller	2018	Rp12000	Tersedia
F016	Midsommar	Thriller	2019	Rp12000	Tersedia
F017	Super Mario Bros Movie	Comedy	2023	Rp10000	Tersedia
F018	Puss in Boots 2	Comedy	2022	Rp10000	Tersedia
F019	How To Train Your Dragon	Animation	2014	Rp10000	Tersedia
F020	Spirited Away	Animation	2001	Rp10000	Tersedia
F021	My Neighbor Totoro	Comedy	1998	Rp12000	Tersedia

[Tekan Enter...]

Gambar 4.22 Hasil Penambahan

```

Pilihan : 6
=== EDIT FILM ===
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Harga/Hr |
+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Rp12000 |
| F002 | Hereditary | Horror | 2018 | Rp12000 |
| F003 | It (2017) | Horror | 2017 | Rp12000 |
| F004 | Talk to Me | Horror | 2023 | Rp12000 |
| F005 | Your Name | Romance | 2016 | Rp10000 |
| F006 | To All The Boys | Romance | 2018 | Rp10000 |
| F007 | Five Feet Apart | Romance | 2019 | Rp10000 |
| F008 | A Silent Voice | Romance | 2016 | Rp10000 |
| F009 | The Dark Knight | Action | 2008 | Rp13000 |
| F010 | John Wick | Action | 2014 | Rp13000 |
| F011 | Mad Max: Fury Road | Action | 2015 | Rp13000 |
| F012 | Demon Slayer: Mugen | Action | 2020 | Rp13000 |
| F013 | Saltburn | Thriller | 2023 | Rp12000 |
| F014 | Gone Girl | Thriller | 2014 | Rp12000 |
| F015 | Bird Box | Thriller | 2018 | Rp12000 |
| F016 | Midsommar | Thriller | 2019 | Rp12000 |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Rp10000 |
| F018 | Puss in Boots 2 | Comedy | 2022 | Rp10000 |
| F019 | How To Train Your Dragon | Animation | 2014 | Rp10000 |
| F020 | Spirited Away | Animation | 2001 | Rp10000 |
| F021 | My Neighbor Totoro | Comedy | 1998 | Rp12000 |
+-----+-----+-----+-----+
Kode film (Tekan Enter -> kembali) : F021
[1]Judul [2]Genre [3]Tahun [4]Harga [5]Studio [6]Negara
Pilihan : 3
Tahun baru : 1988
[v] Film diedit!
[Tekan Enter...]=====

```

Gambar 4.23 Menu Admin Pilihan 6

```

Pilihan : 1
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Rp12000 | Tersedia |
| F002 | Hereditary | Horror | 2018 | Rp12000 | Tersedia |
| F003 | It (2017) | Horror | 2017 | Rp12000 | Tersedia |
| F004 | Talk to Me | Horror | 2023 | Rp12000 | Tersedia |
| F005 | Your Name | Romance | 2016 | Rp10000 | Tersedia |
| F006 | To All The Boys | Romance | 2018 | Rp10000 | Tersedia |
| F007 | Five Feet Apart | Romance | 2019 | Rp10000 | Tersedia |
| F008 | A Silent Voice | Romance | 2016 | Rp10000 | Tersedia |
| F009 | The Dark Knight | Action | 2008 | Rp13000 | Tersedia |
| F010 | John Wick | Action | 2014 | Rp13000 | Tersedia |
| F011 | Mad Max: Fury Road | Action | 2015 | Rp13000 | Tersedia |
| F012 | Demon Slayer: Mugen | Action | 2020 | Rp13000 | Tersedia |
| F013 | Saltburn | Thriller | 2023 | Rp12000 | Tersedia |
| F014 | Gone Girl | Thriller | 2014 | Rp12000 | Tersedia |
| F015 | Bird Box | Thriller | 2018 | Rp12000 | Tersedia |
| F016 | Midsommar | Thriller | 2019 | Rp12000 | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Rp10000 | Tersedia |
| F018 | Puss in Boots 2 | Comedy | 2022 | Rp10000 | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014 | Rp10000 | Tersedia |
| F020 | Spirited Away | Animation | 2001 | Rp10000 | Tersedia |
| F021 | My Neighbor Totoro | Comedy | 1988 | Rp12000 | Tersedia |
+-----+-----+-----+-----+-----+
[Tekan Enter...]

```

Gambar 4.24 Hasil Pengeditan

```

Pilihan : 7
=== HAPUS FILM ===
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Status |
+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Tersedia |
| F002 | Hereditary | Horror | 2018 | Tersedia |
| F003 | It (2017) | Horror | 2017 | Tersedia |
| F004 | Talk to Me | Horror | 2023 | Tersedia |
| F005 | Your Name | Romance | 2016 | Tersedia |
| F006 | To All The Boys | Romance | 2018 | Tersedia |
| F007 | Five Feet Apart | Romance | 2019 | Tersedia |
| F008 | A Silent Voice | Romance | 2016 | Tersedia |
| F009 | The Dark Knight | Action | 2008 | Tersedia |
| F010 | John Wick | Action | 2014 | Tersedia |
| F011 | Mad Max: Fury Road | Action | 2015 | Tersedia |
| F012 | Demon Slayer: Mugen | Action | 2020 | Tersedia |
| F013 | Saltburn | Thriller | 2023 | Tersedia |
| F014 | Gone Girl | Thriller | 2014 | Tersedia |
| F015 | Bird Box | Thriller | 2018 | Tersedia |
| F016 | Midsommar | Thriller | 2019 | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Tersedia |
| F018 | Puss in Boots 2 | Comedy | 2022 | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014 | Tersedia |
| F020 | Spirited Away | Animation | 2001 | Tersedia |
| F021 | My Neighbor Totoro | Comedy | 1988 | Tersedia |
+-----+-----+-----+-----+
Kode film (Tekan Enter -> kembali) : F021
Hapus "My Neighbor Totoro"? (y/n) : y
[v] Film dihapus!
[Tekan Enter...]

```

Gambar 4.25 Menu Admin Pilihan 7

```

Pilihan : 1
+-----+-----+-----+-----+-----+
| Kode | Judul | Genre | Tahun | Harga/Hr | Status |
+-----+-----+-----+-----+-----+
| F001 | The Nun II | Horror | 2023 | Rp12000 | Tersedia |
| F002 | Hereditary | Horror | 2018 | Rp12000 | Tersedia |
| F003 | It (2017) | Horror | 2017 | Rp12000 | Tersedia |
| F004 | Talk to Me | Horror | 2023 | Rp12000 | Tersedia |
| F005 | Your Name | Romance | 2016 | Rp10000 | Tersedia |
| F006 | To All The Boys | Romance | 2018 | Rp10000 | Tersedia |
| F007 | Five Feet Apart | Romance | 2019 | Rp10000 | Tersedia |
| F008 | A Silent Voice | Romance | 2016 | Rp10000 | Tersedia |
| F009 | The Dark Knight | Action | 2008 | Rp13000 | Tersedia |
| F010 | John Wick | Action | 2014 | Rp13000 | Tersedia |
| F011 | Mad Max: Fury Road | Action | 2015 | Rp13000 | Tersedia |
| F012 | Demon Slayer: Mugen | Action | 2020 | Rp13000 | Tersedia |
| F013 | Saltburn | Thriller | 2023 | Rp12000 | Tersedia |
| F014 | Gone Girl | Thriller | 2014 | Rp12000 | Tersedia |
| F015 | Bird Box | Thriller | 2018 | Rp12000 | Tersedia |
| F016 | Midsommar | Thriller | 2019 | Rp12000 | Tersedia |
| F017 | Super Mario Bros Movie | Comedy | 2023 | Rp10000 | Tersedia |
| F018 | Puss in Boots 2 | Comedy | 2022 | Rp10000 | Tersedia |
| F019 | How To Train Your Dragon | Animation | 2014 | Rp10000 | Tersedia |
| F020 | Spirited Away | Animation | 2001 | Rp10000 | Tersedia |
+-----+-----+-----+-----+-----+
[Tekan Enter...]

```

Gambar 4.26 Hasil Penghapusan

```

Pilihan : 8
=== SEMUA RIWAYAT SEWA ===
+-----+-----+-----+-----+-----+
| ID | Penyewa | Film | Total | Status |
+-----+-----+-----+-----+-----+
| 1 | Nurul | Demon Slayer: Mugen | Rp83000 | Selesai |
+-----+-----+-----+-----+-----+
[Tekan Enter...]

```

Gambar 4.27 Menu Admin Pilihan 8

```
=====
MENU ADMIN | Admin
1. Lihat Film
2. Sewa Film
3. Kembalikan
4. Riwayat Sewa
-----
5. Tambah Film
6. Edit Film
7. Hapus Film
8. Semua Riwayat
0. Logout
-----
Pilihan : 0
Sampai jumpa, Admin!
[Tekan Enter...]
```

Gambar 4.28 Menu User dan Admin Pilihan 0

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\Users\LOQ\Documents\praktikum-apl> git add .
```

Gambar 5.1 Git Add

Digunakan untuk menambahkan perubahan file ke staging area (area persiapan).

5.2 GIT Commit

```
PS C:\Users\LOQ\Documents\praktikum-apl> git commit -m "Codingan Post test 6"
[main 684275f] Codingan Post test 6
4 files changed, 1049 insertions(+)
create mode 100644 A2'25/Pertemuan-6/Pertemuan-6.cpp
create mode 100644 A2'25/Pertemuan-6/Pertemuan-6.exe
create mode 100644 post-test/post-test-apl-6/2509106033_NurulLatifa_PT-6.cpp
create mode 100644 post-test/post-test-apl-6/2509106033_NurulLatifa_PT-6.exe
```

Gambar 5.2 Git Commit

Digunakan untuk menyimpan perubahan yang sudah di-add ke dalam repository lokal dengan pesan tertentu. (ini tadi ada salah penamaan commit, mohon maaf)

5.3 GIT Push

```
PS C:\Users\LOQ\Documents\praktikum-apl> git push
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 20 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 123.91 KiB | 5.16 MiB/s, done.
Total 10 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:  https://github.com/nulaatifa/praktikum-apl.git
To https://github.com/nulaatifa/Praktikum_Algoritma_Pemograman_Lanjut.git
5d2fb56..684275f  main -> main
PS C:\Users\LOQ\Documents\praktikum-apl>
```

Gambar 5.3 Git Push

Digunakan untuk mengirim hasil commit dari repository lokal ke repository remote (misalnya ke GitHub).